

# The diversity of high-frequency traders

Björn Hagströmer\* & Lars Nordén

Stockholm University School of Business, S-106 91 Stockholm

May 18, 2013

## Abstract

The regulatory debate concerning high-frequency trading (HFT) emphasizes the importance of distinguishing different HFT strategies and their influence on market quality. Using data from NASDAQ-OMX Stockholm, we compare market-making HFTs to opportunistic HFTs. We find that market makers constitute the lion's share of HFT trading volume (63-72%) and limit order traffic (81-86%). Furthermore, market makers have higher order-to-trade ratios and lower latency than opportunistic HFTs. In a natural experiment based on tick size changes, we find that the activity of market-making HFTs mitigates intraday price volatility.

*Key words:* High-frequency trading; Market making; Market quality; Volatility

*JEL codes:* G14, G18

---

\*Corresponding author. Phone: + 46 8 163030; Fax: +46 8 6747440; E-mail: [bjh@fek.su.se](mailto:bjh@fek.su.se). Lars Nordén e-mail: [ln@fek.su.se](mailto:ln@fek.su.se)

## 1 INTRODUCTION

Recent advances in information technology employed in equity markets allow traders to process information and submit orders at lightning speed. With typical holding periods measured in seconds or even fractions of seconds, resulting in large trading volumes, algorithmic strategies are now major forces in equity markets. Since the arrival of such strategies has coincided with massively increased limit order submissions and cancellations, high intraday price volatility (including flash crashes), and fragmentation of volumes across marketplaces, many voices have been raised, calling for the regulation of algorithmic trading (AT).

To understand the influence of AT on the quality of markets, it is important to realize that investors use algorithms for many different reasons and strategies. The mere fact that a strategy is algorithmic or based on fast trading does not determine its market impact. To distinguish the main strategies and their individual market impacts is thus pivotal for effective regulatory design. A useful distinction to make is to separate agency algorithms from proprietary algorithms. The former is typically a service provided to clients to minimize the price impact of trading. Proprietary algorithms, on the other hand, are used by technologically sophisticated firms aiming to profit from the trading process itself. Firms using proprietary algorithms are referred to as high-frequency traders (HFTs).<sup>1</sup> HFT strategies may be further subdivided into market-making strategies, on the one hand, and opportunistic trading such as arbitrage and directional (order anticipation and momentum ignition) trading, on the other (SEC, 2010).

As limit order book markets in general preserve trader anonymity, empirical analysis of distinct AT and HFT strategies is difficult. Academic research in this field is based on

---

<sup>1</sup> We use *HFT* as an abbreviation for both *high-frequency trader* and *high-frequency trading*.

market-wide proxies of algorithmic activities (e.g., Boehmer et al., 2012; Hendershott et al., 2011) or proprietary data sets (e.g., Brogaard et al., 2012; Hirschey, 2011; Kirilenko et al., 2011). None of these papers has been able to distinguish different HFT strategies in equity markets.

In this paper, we use a proprietary data set that allows us to observe all limit order submissions, cancellations, and executions, complete with the identities of the traders.<sup>2</sup> Our sample includes thirty Swedish large-cap stocks traded on NASDAQ-OMX Stockholm exchange (henceforth NOMX-St). A key contribution of our paper is that, by using detailed trading and quoting information, we are able to subcategorize HFTs into market-making strategies, and opportunistic (e.g., arbitrage and directional) strategies. Our categorization methodology, combined with rich order book information, allows us to conduct a more detailed analysis of HFT in equity markets than previous studies managed. We carry out the analysis for one highly volatile month (August, 2011) and one relatively calm month (February, 2012).

We define market-making HFTs as proprietary-only firms that use algorithms in their order submission, and that have a continuous presence at the best bid and offer prices in the limit order book. We find that, within the group of HFTs, such market makers represent around 71.5% of the trading volume in August 2011, and 62.8% in February 2012. During both months, more than 80% of the HFT limit order submissions originate from the market-making strategies. The market-making HFTs in our study conform to the market makers described by Jovanovic and Menkveld (2012) and Menkveld (2013), in that they trade large volumes but keep inventories close to zero, and in that they are

---

<sup>2</sup> The results of this paper are aggregated and do not reveal specific trader strategies. Furthermore, the data showing trader identities was never removed from NASDAQ OMX's possession. All trader specific analysis was performed on site.

on the passive side in a majority of their trades. The result that is new to the literature is that market-making HFTs have higher order-to-trade ratios and lower latency than opportunistic HFTs. This reflects modern market making, where anyone who is unable to respond immediately to news about fundamentals or the order flow, by modifying posted orders, will be picked off by competitors.

As a second contribution of this paper, we study the influence of market-making HFTs on short-term volatility. To investigate causality between HFT activity and market quality, it is necessary to overcome the endogeneity problem posed by the fact that HFTs are selectively active in order books where market quality is high. In previous literature, this problem has been addressed through the use of exogenous events that influence AT or HFT activity, but do not directly influence market quality. For example, Hendershott et al. (2011) use the automation of quotes on the New York Stock Exchange, and Boehmer et al. (2012) use the availability of co-location services to proxy for AT activity. Both find that AT has a positive influence on liquidity, a finding also supported by Brogaard (2011; for HFTs), Hasbrouck and Saar (2013), and Hendershott and Riordan (2012). Brogaard et al. (2012) and Hendershott and Riordan (2012) find that HFT and AT, respectively, contribute to price discovery. As pointed out by Boehmer et al. (2012), however, there is less agreement on how AT and HFT influence volatility. Whereas Hasbrouck and Saar (2013) find a positive effect, Kirilenko et al. (2011) find that HFTs may have worsened the flash crash in May 2010, and Boehmer et al. (2012) find that short-term volatility is amplified by AT.

We use tick size changes as exogenous events to study HFT's influence on short-term volatility. On European stock exchanges, the tick size (minimum price increment) depends on the stock price level. For example, when the price of a stock increases from

SEK 49 to SEK 51, the tick size increases fivefold from SEK 0.01 to SEK 0.05 (SEK is the abbreviation for the Swedish currency *krona*). We hypothesize that an increased tick size makes market making more profitable, and other strategies, such as arbitrage trading, more costly. This is because market makers typically earn the spread, whereas opportunistic traders tend to pay the spread. Thus, we predict that a tick size increase would increase the share of trading activity due to market-making HFTs, whereas the share of trading activity due to opportunistic HFTs' activities would decrease. A reduction in the minimum tick size would have the opposite effect. Our results, which in this application are based on more than two years of data (February 8, 2010 to March 31, 2012), confirm that this is indeed the case. When using the tick size events to study the influence of HFT on short-term volatility, we find that an increase in market-making HFT activity mitigates short-term volatility.

This paper is closely related to that of Brogaard (2011) and Carrion (2013). They both investigate HFT activity in a set of 120 US stocks trading on the NASDAQ, using an indicator provided by the exchange to determine whether or not a transaction is due to HFT. While Brogaard (2011) focuses his analysis of characteristics of HFTs' trading activities, Carrion (2013) relates HFTs' activities to stock market efficiency. A distinct feature of our paper compared to their papers is our ability to distinguish different types of HFT. Our finding that HFTs are indeed a heterogeneous group of traders shows the importance of distinguishing HFT strategies. Hirschey (2011) uses the same HFT classification as Brogaard (2011) and Carrion (2013) and, in addition, observes trader identities. In his study on anticipatory trading, he finds that HFTs differ in their ability to forecast future order flows, but he does not relate such abilities to differences in HFT strategies.

Kirilenko et al. (2011) are able to observe the trades of individual firms and investigate the trading and quoting surrounding the market turbulence of May 6, 2010, known as the flash crash. They define HFTs as the 7% most active intermediaries in the market, and find that HFTs did not cause, but may have amplified, the volatility in the flash crash. Their data concern only one security, the S&P 500 E-mini futures, over three days of extraordinary volatility. Our investigation can be seen as a complement to their paper, as we cover thirty stocks over two months of different volatility levels.

Finally, our paper relates to the work of Jovanovic and Menkveld (2012). They develop a model of market-making HFT, showing that the presence of such traders may either increase or reduce adverse selection costs, depending on how well-informed liquidity demanders are. In an empirical application, they show that one HFT that dominated the trading on the Chi-X in 2007-2008 behaved as a middleman in their model, with low net inventories, predominantly passive trades, and fast trading. The HFT market makers investigated in this paper have the same properties.

In the next section, we present our empirical setting and data, as well as our HFT categorization methodology. After that, we present estimates of various metrics frequently associated with HFT. In Section 4, we divide the HFT group into subcategories. Specifically, we study how market makers differ from other HFTs. Furthermore, we investigate how the behavior of market makers and opportunistic traders differs across segments of stocks, and how market-making activity is related to market quality measures. In Section 5, we present our event study, where we analyze the causal effects of HFT activity on volatility. Section 6 offers concluding remarks.

## 2 INSTITUTIONAL DETAILS, DATA, AND HFT CLASSIFICATION

We study stock trading on NOMX-St, which is an electronic limit order book market that operates continuous trading from 9 am to 5.30 pm every weekday, except on Swedish bank holidays. If the day before a Swedish bank holiday is a weekday, trading on that day closes at 1 pm. Opening and closing prices are determined in call auctions.

NOMX-St uses the INET trading system to handle submissions and cancellations of limit orders. Limit orders convey a trading interest for a given quantity at a limit price, valid for a specified time period. Execution during the continuous trading session takes place when two limit orders cross, following an order of priority by price, member, visibility, and time.<sup>3</sup> Non-executed limit orders may be partially or completely cancelled at any time. Price modification orders, which allow traders to change the price of a limit order, were not available during the period of our sample.

As is common in electronic limit order book markets, limit orders may be entered with various conditions imposed. At NOMX-St, limit orders may be either partially or completely non-visible. Complete non-visibility is only available for orders whose value exceeds EUR 50,000 (or more, depending on daily turnover of the stock). Due to this rule, partially and completely hidden orders constitute no more than 0.7% of all limit orders in our sample.<sup>4</sup> Limit orders that are conditioned to be cancelled immediately if they are not executable may also specify a minimum quantity for execution. Finally, the limit order price can be pegged to the same side, the opposite side, or the midpoint of

---

<sup>3</sup> The "member" priority rule implies that a member with a limit order posted at the best price has execution priority if a market order originating from the same member is posted on the other side of the book.

<sup>4</sup> As a point of comparison, Bessembinder et al. (2009) report that, in a sample of 100 stocks on Euronext Paris, gathered in April 2003, 44% of the order volume is hidden. In a sample of 99 NASDAQ stocks from October 2010, Hautsch and Huang (2012) report that hidden liquidity is associated with "enormous order activities" related to liquidity detection strategies (p.2). The lack of hidden liquidity on NOMX-St is likely to induce fewer liquidity detection strategies, which in general should lead to lower order-to-trade ratios.

the prevailing best bid and offer (BBO) prices, implying that the limit order price is automatically adjusted whenever the BBO is changed (with a loss of time priority).

NOMX-St has roughly one hundred member firms that have the right to submit orders to trade on the stock market. Exchange members may sell trading services to traders either as traditional brokers, through direct market access, or through sponsored access. Direct market access gives customers access to the market through the infrastructure of the member firm. In the case of sponsored access, the customer uses its own infrastructure but trades under the member identity of the sponsor. Sponsored access is becoming increasingly popular among AT firms, particularly HFTs, as it allows for lower latency (order-processing time) than direct market access.

Around 2009-2010, NOMX-St introduced several changes in the trading system to cut both the cost and the latency of trading. The changes included central counterparty clearing (October 2009), capped trading fees (January 2010), and the INET trading platform (February 2010). All these changes facilitate the activity of HFTs.

The INET trading system has the capacity to handle more than a million messages per second, at an average processing time of less than 0.25 milliseconds. Brokers' own systems add additional latency due to network and processing delays. To cut network latency (order-processing time) further, NOMX-St offers co-location services, through which member firms and sponsored access subscribers can pay a premium to place their servers on the premises of the exchange.

NOMX-St applies a differentiated price list for liquidity makers and liquidity takers. Limit orders that stay in the order book until they are hit by an incoming order are seen as maker orders, while limit orders that lead to immediate execution are taker orders. In

every trade, the trader behind the maker order pays a lower fee than the trader behind the taker order. For the time period analyzed here, NOMX-St did not offer rebates (negative fees) to any market participants. The capped trading fees mentioned above, however, may bring the marginal cost of trading down to zero, as they put a ceiling on the nominal amount of fees charged in a given month.

## **2.1 Data and sample selection**

We access all messages that are available in the data feed ITCH. ITCH includes all limit orders, cancellations and executions, but is somewhat restricted in terms of the information it provides on limit order conditions.<sup>5</sup> For each limit order submission, we observe the time of entry, the quantity, the limit price, trader identity information, visibility conditions, and time in force. Whenever a limit order is cancelled or executed (partially or completely), a new entry is made in ITCH. To track the life of a limit order, ITCH assigns an order sequence number that connects submissions, cancellations and executions. For each trade, ITCH creates two execution messages, one for the maker and one for the taker. A matching number is assigned to each execution message to keep track of which messages belong together.

The trader identity information associated with each ITCH message is of high importance for this study. We utilize publicly available market participant identity (MPID) information as well as exchange level information indicating whether an order is automated. Automated orders are eligible for a fee discount.

Our sample is limited to the thirty most-traded stocks that together constitute the OMXS 30, a leading Swedish stock index. We limit our sample to the OMXS 30 constituents

---

<sup>5</sup> For example, partially hidden orders (icebergs) are reported in ITCH as two limit orders, one visible and one hidden. Whenever a part of the hidden volume is uncovered, it is entered in ITCH as a cancellation of the hidden volume and the submission of a new limit order. Pegged orders follow the same logic.

because HFTs tend to concentrate their activity on the most-traded stocks. We access all trading activity from February 8, 2010 to March 31, 2012. The first date marks the introduction of the INET low-latency trading platform, and the end date is the last trading day before a new liquidity program was introduced, when the trading costs for market makers were lowered. An inherent problem with rich market microstructure data sets is the computational burden. For that reason, we limit most of the analysis in this article to just two months of trading, August 2011 and February 2012. August 2011 was a highly volatile month, in which HFT attracted extensive media attention. Reasons for the volatility include worries about credit rating downgrades of the United States and France, as well as concerns about the sovereign debt crisis spreading to Italy and Spain. Being a much less volatile month, February 2012 is included for comparison.

NOMX-St hosts 55-65% of the trading volume in OMXS 30 stocks. The largest competitor is BATS Chi-X Europe (25-30%), followed by Burgundy and Turquoise (less than 5% each).<sup>6</sup> Our data set is limited to trading and quoting activity on NOMX-St. As a final limitation, only messages from the continuous trading session are included, representing roughly 97% of all trades and 99% of all limit order submissions.

Having access to all messages in the ITCH database, we are able to reconstruct the state of the order book at any given point in time during continuous trading. As a complement to the ITCH data set, we access data on stock market capitalization from the NOMX-St website. Furthermore, to study prices, trading volumes and realized volatility on days where we do not analyze HFT trading (see Sections 4.3 and 5.1), we retrieve information on prices and volumes at a 15-minute frequency from Thomson Reuters' Tick History database, maintained by the Securities Research Centre of Asia-Pacific (SIRCA).

---

<sup>6</sup> Market share data are taken from [http://www.batstrading.co.uk/market\\_data/venue/index/OMXS/](http://www.batstrading.co.uk/market_data/venue/index/OMXS/)

## 2.2 Descriptive statistics

Table 1 presents summary statistics. The first three rows include means, medians, and standard deviations estimated for August 2011, and the middle three rows the same statistics for February 2012. The bottom three rows contain the p-values of the tests for differences, where each null hypothesis states that there is no difference in the statistic in question between the two months.

The average market capitalization of the stocks in our sample is SEK 79 billion in August 2011, and SEK 91 billion in February 2012, corresponding to USD 12 billion and USD 14 billion at an exchange rate of SEK/USD 6.6. Thus, the stocks analyzed here are slightly smaller on average than those analyzed in the HFT study by Brogaard et al. (2012), which averaged USD 18 billion. The average relative bid-ask spread in our sample (0.09-0.10%) lies between those of their medium-cap and large-cap stocks (0.05% and 0.13%).

It is clear from Table 1 that our two sample months differ substantially from one another. The mean realized one-minute volatility is 0.215 in August 2011, but only 0.044 in February 2012 (the difference being statistically significant both in terms of means and medians).<sup>7</sup> August 2011 also records significantly less liquidity in terms of depth, nominal bid-ask spreads, and relative bid-ask spreads (the liquidity measures are based on limit order book snapshots taken every minute for each stock). The mean number of trades is 52% higher in August 2011 than in February 2012, and the mean number of limit order submissions (cancellations) is 318% (311%) higher. Given the large differences, we report results for the two months separately in our subsequent analysis.

It is notable from Table 1 that the number of cancellations is higher than the number of

---

<sup>7</sup> The realized one-minute stock return volatility is defined as the squared midpoint quote change in the one-minute interval.

limit order submissions. This is because the volume of limit orders is often partially cancelled. For example, a limit order stating a trading interest of ten shares may be subject to up to ten different executions and cancellations.

### **2.3 HFT and non-HFT trader classification**

The use of algorithms in trading is widespread nowadays. In a survey conducted by the Swedish Financial Supervisory Authority, 20 (7 banks and 13 institutional investors) out of the 24 financial firms that participated claimed that they use algorithms in their trading (Finansinspektionen, 2012). In order to investigate the nature and impact of AT it is thus necessary to break the activity down into subcategories. The categorization of traders is, in general, complicated by the fact that traders do not tend to stick to any one strategy. On the contrary, they adapt and change their strategies in accordance with their expected returns and risk-taking preferences. An important distinction, however, is whether traders apply their strategies to their own holdings or as services to clients. Applying this distinction to AT, we get two subgroups of AT, agency algorithms and proprietary algorithms, where the latter is what is typically referred to as HFT. Agency algorithm firms provide execution services for clients, typically using their infrastructure and market knowledge to minimize the price impacts of trading. HFT strategies, on the other hand, may be further subdivided into market-making and opportunistic trading, where the latter includes strategies such as arbitrage and directional (order anticipation and momentum ignition) trading.<sup>8</sup>

In order to analyze the behavior of HFTs on the NOMX-St, we classify market member firms into three categories: (1) members that are primarily HFTs, i.e., that engage in proprietary trading only and use algorithms in their trading strategies; (2) members

---

<sup>8</sup> See SEC (2010) and Gomber et al. (2011) for definitions and details about these strategies.

that primarily trade for clients; and (3) members that engage in both proprietary and client trading. The categorization is done with the aid of NASDAQ OMX in-house expertise about member activities. As a result, 29 MPIDs are identified as HFT firms, 49 as non-HFT firms, and 22 as hybrid firms that engage in both proprietary and agency activities.<sup>9</sup> Our data set does not allow us to isolate the activities of HFTs that access the market through sponsored access. Thus, such activity will end up among the hybrid firms.

To confirm that the HFT firms are indeed using algorithms in their proprietary trading, we filter their activity with respect to automated order information. Messages identified as automated are required by the exchange to be algorithmic. We find that 98.2% of all messages (96.7% of the trading volume) entered by members in the HFT group originate from algorithmic orders. The high percentage of such orders within the group is taken as proof of the validity of the qualitative categorization process. In the non-HFT group and the hybrid group, respectively 13.5% and 63.6% of the messages (14.0% and 53.0% of the trading volume) originate from algorithmic orders, confirming that the use of AT is widespread in the market.

Our HFT categorization is similar to the procedure applied to a data set of NASDAQ stocks by Brogaard (2011), Brogaard et al. (2012), Carrion (2013), and Hirschey (2011). The HFT categorization used for the US data set is richer in the sense that it is dynamic (continuously updated) whereas ours is done at one point in time (May 2012). One advantage of our categorization relative to that of the US data set is that, along with the HFT group, we also identify a group that is free from HFT activities. This feature allows

---

<sup>9</sup> Confidentiality requirements do not allow us to disclose the members in each category, but the complete member list is publicly available online at <http://nordic.nasdaqomxtrader.com/membershipservices/membershiplist/>

us to benchmark our HFT results against a group of non-HFT members. Furthermore, whereas the US data set contains quote information at BBO prices only, we access order information at all levels in the order book. Having full information on the limit order book enables a richer analysis of order submission strategies, though the picture is incomplete in the sense that we do not observe activity on other exchanges. The key advantage of our data set, however, is that we are able to observe the activity of individual HFT members, whereas the US data set only indicates whether a given trade or quote is associated with one of the 26 HFT firms but not which one.

Trader categorizations based on trading activity impose priors on what traders actually do. Our HFT categorization is conditional only on the use of algorithms for proprietary trading. This enables us to assess whether actual HFT trading activity is in line with common conceptions thereof, which we do in the next section.

### **3 CHARACTERISTICS OF HIGH-FREQUENCY TRADERS**

Given the plethora of HFT definitions in the current literature (see Gomber et al., 2011), it is interesting to study several dimensions of HFT activity. According to SEC (2010), HFTs tend to (i) end the day with close to zero inventories; (ii) frequently submit and cancel limit orders; (iii) use co-location facilities and highly efficient algorithms, allowing them to minimize different types of latencies; and (iv) have short holding periods. The aim of our analysis in this section is to determine whether such conceptions of HFT behavior apply to our sample of HFT members.

Along with measures of trading and quoting volumes, we report various metrics for the HFT sample as well as the control group of non-HFT members. The group of residual (hybrid) members is likely to contain substantial HFT activity, entered through, e.g.,

sponsored access or the trading desks of banks that also have clients. As this group is a mixture of HFTs and non-HFTs, and any results related to its activities would be difficult to interpret, we do not explicitly analyze its activities.

### 3.1 Metrics of trading activity

Our first metric of HFT behavior is the absolute day-end inventory (the sum of all signed trading volumes) divided by daily trading volume, denoted  $|Inventory|$ . To retrieve an aggregate daily measure, we report the average  $|Inventory|$  across members and stocks for each trading day. Market makers and opportunistic traders alike strive to have low  $|Inventory|$  as positions held overnight are subject to clearing and capital costs. However, as we are unable to observe trades on other exchanges, this metric may falsely indicate a high inventory for firms that in fact utilize central clearing for trades on several exchanges (e.g., cross-market arbitrage strategies) or several asset classes. Hence, a high  $|Inventory|$  may be seen as an indication of arbitrage activities by an HFT.

A second feature often associated with HFT is the frequent submissions and cancellations of limit orders. As the algorithms continuously scan the markets for news about fundamentals, order flows, and related prices, the optimal quotes are subject to continuous change. The most common metric for quoting intensity is the order-to-trade ratio,  $q/t$ . We define  $q/t$  as the number of limit order submissions across members and stocks on a given trading day, divided by the number of executions across members and stocks on that day. Hendershott et al. (2011) use a related metric, the number of messages (limit order submissions and cancellations) per \$100 of trading volume, as a proxy for market-wide AT activity. Boehmer et al. (2012) use two alternative measures of quoting intensity to approximate the amount of AT.

A dimension of trading activity that is closely related to HFT is the speed of trading. We measure the lifetime of each limit order as the time that elapses from limit order submission to its first cancellation (limit order volume can be partially cancelled). Our measure of *Minimum latency* is the minimum limit order lifetime for each member, each day, and each stock. Furthermore, we calculate the average limit order lifetime for each member, stock, and day, as a measure of *Limit order duration*. To retrieve daily measures of *Minimum latency* and *Limit order duration*, we report the median across members and stocks on each trading day. Hasbrouck and Saar (2013) argue that low-latency trading strategies are a hallmark of HFTs, and show that HFTs on the US NASDAQ are able to respond to news events within 2-3 milliseconds (based on a sample from 2007-2008). Several theoretical models use latency as the defining characteristic of AT and HFT, finding that fast traders profit at the expense of slow traders (Jarrow and Protter, 2011; McNish and Upson, 2011). According to Biais et al. (2011), such potential profits lead HFTs to overinvest in technology.

We are also interested in how often members of different groups act as liquidity suppliers, i.e., how often their limit orders are hit by market orders. For this purpose, we define *Liquidity supply* as the fraction of trades, across members and stocks for each trading day, where the trader acts as a liquidity supplier. We report this metric in terms of both SEK trading volume and number of trades. Similar metrics are reported by Kirilenko et al. (2011) and Jovanovic and Menkveld (2013).

### **3.2 HFT and non-HFT trading activity**

Table 2 displays trading and order volumes as well as the metrics described above, calculated for the HFTs and the non-HFTs. Results for August 2011 are given in Panel A and results for February 2012 in Panel B. The bottom three rows of each panel contain

the p-values of the tests for differences in mean, median, and standard deviation between the HFTs and the non-HFTs. The statistics are reported across trading days, so that the number of observations for the statistical tests is 23 for August 2011, and 21 for February 2012.

In August 2011, the HFTs represent on average about 29% of the total trading volume in the OMXS 30 stocks. The fraction is a little lower in February 2012, at 26%. Both figures are much lower than the almost 70% that Brogaard (2011) and Carrion (2013) report for US markets (Brogaard, 2011, for NASDAQ and BATS, and Carrion, 2013, for NASDAQ). Any comparison of such numbers is, however, complicated by the fact that hybrid member firms (with both HFT and agency services) are excluded. Hybrid members represent more than 40% of the trading volume, and more than 60% of the limit order submissions, in both August 2011 and February 2012. Given that 53% of the hybrid firms' trading volume is entered using algorithmic orders (see Section 2.3), we conclude that the HFT activity in the OMXS 30 stocks ranges between 25% and 50% of the total activity.

Interestingly, the share of limit order submissions traced to HFTs is on par with the trading activity, at around 30%. This indicates that the HFTs in our sample do not overflow the market with limit orders at a rate higher than average. Still,  $q/t$  is significantly higher for the HFTs than for the non-HFTs (in terms of both mean and median). For both trader groups,  $q/t$  is slightly higher in the more volatile month (August 2011). The results in Table 2 imply that the hybrid traders have an average  $q/t$  ratio that is higher than the HFTs. This is an indication of order-intensive strategies within the hybrid group that may well be high-frequency strategies. As discussed above,

we are unable to distinguish that activity, which may come from sponsored access clients or proprietary trading desks at commercial banks, from client orders.

The two ratios indicating the liquidity supply show that the HFTs are more passive than the non-HFTs. The differences are statistically significant for both months, and for both means and medians. This is a sign of market-making activity among HFTs. As all trades have one passive and one active party, the average ratio across all traders should be equal to unity. The ratios reported indicate that the HFTs supply more liquidity than they demand, that the non-HFTs supply roughly as much as they demand, and accordingly that the third group of traders (hybrids) are net demanders of liquidity. The HFT group's liquidity supply ratios (around 57-59%) are much lower than the figures of 74-79% observed by Jovanovic and Menkveld (2012) for one HFT market maker trading Dutch stocks on the Chi-X. Our results are, however, on average slightly higher than those of Kirilenko et al. (2011), who report trade-weighted (volume-weighted) liquidity supply ratios of around 50% (54%) for their HFTs.

The */Inventory/* metric shows that the HFTs, in both months, close the trading day with less net inventory than the non-HFTs. This is in line with expectations, as HFTs by definition are proprietary traders that are well-known for minimizing their overnight exposure. Jovanovic and Menkveld (2012) report that their Chi-X market maker closes with zero inventory on 33-60% of the trading days in their sample period. The fact that */Inventory/* is non-zero on average for the HFTs may indicate that their strategies span several order books, as would be the case with arbitrage strategies.

Finally, our investigation shows that the HFTs are faster than their non-HFT peers. Both the minimum latency and the average order lifetime are significantly lower for the HFTs. According to theory (see, e.g., Biais et al., 2011), the ability to adapt quickly puts HFTs at

an informational advantage, allowing them to extract an adverse selection cost from slower traders. For both trader groups in our sample, the *Minimum latency* is lower in February 2012 than in August 2011. This may reflect an improvement in technology over time. The latency gap between HFTs and non-HFTs decreases but remains statistically significant in February 2012. Furthermore, for both trader groups, the *Limit order duration* is longer in February 2012 than in August 2011. This effect is likely to be due to the lower trading activity and lower volatility seen in February 2012.

## 4 DISTINGUISHING MARKET MAKERS FROM OTHER HFTS

So far, all our results have been reported for the HFTs and non-HFTs as whole groups. This categorization is in line with the current HFT literature, enabling an analysis of the aggregate effects of HFT behavior. As our data set allows us to observe the behavior of each member firm separately, we now turn to disaggregating the HFT group (for confidentiality reasons, we still aggregate our results at the member group level). As discussed in the introduction, such disaggregation is encouraged in the regulatory debate. In this section, we introduce a measure of market-making activity that allows us to separate HFT market makers from opportunistic HFTs. We investigate how market-making and opportunistic HFT activities differ from each other, in general and in various market segments.

### 4.1 Measuring market-maker presence

To investigate the degree of market making among HFTs, we study the prevalence of each HFT member firm at the inside quotes. For each stock, we take snapshots of the order book for each ten-second period on each trading day during each of our two sample months. As algorithms may post limit orders cyclically, for example at the turn of

minutes or seconds (see Hasbrouck and Saar, 2013), we randomize the order book snapshot time within each ten-second period to get observations that are uniformly distributed across the ten seconds. The *market-maker presence* for each HFT member is then calculated daily for each stock as the fraction of snapshots in which the member has a limit order posted at either side of the inside quotes.

Our measure of market-maker presence is analogous to a measure applied by NOMX-St in a new pricing scheme for market makers introduced in April 2012. Similar measures are used by Brogaard (2011) and Brogaard et al. (2012) but with the important restriction that they measure the *aggregate* presence of HFTs and algorithmic traders, respectively, rather than the presence of individual members.

The results on market-maker presence are presented graphically in Figure 1, with findings for August 2011 in Panel A, and for February 2012 in Panel B. The graph can be interpreted as a three-dimensional bar chart, with OMXS 30 stocks on the x-axis and HFT members on the z-axis. The HFT members are sorted by their average degree of market making across stocks and trading days, and the stocks are presented in random order (names of stocks and members are dropped for confidentiality reasons). The market-maker presence, averaged across the trading days in each month, is displayed on the y-axis.

The most striking result demonstrated by Figure 1 is that market-maker presence in OMXS 30 stocks is concentrated among a handful of HFT members. This group of traders is highly active across the range of stocks in our sample. Other HFT members are present at the inside quotes sporadically, but rarely on the continuous basis that is associated with market making. The concentration of market-making among a few HFTs holds in both August 2011 and February 2012.

As market-maker presence appears to be a property that clearly distinguishes trading strategies, we use it to disaggregate the HFTs. We postulate that members with a more than 20% market-maker presence, on average across the stocks, are likely to have market making as their main business model. Such members are labeled *market-making HFTs*. Members with a lower market-maker presence are likely to employ arbitrage or directional strategies as their main activity, and we label these members *opportunistic HFTs*. In the subsequent analysis in this section, we compare the activities of these two subcategories of HFTs.<sup>10</sup>

## 4.2 Market-making and opportunistic HFT activity

Table 3 provides a comparison of the trading activity of market-making HFTs and opportunistic HFTs. The trading volume and number of limit orders are presented as fractions of all HFT activity. The *Market maker presence*,  $|Inventory|$ ,  $q/t$ , *Minimum latency*, *Limit order duration*, and *Liquidity supply* metrics are defined as above. To conserve space, we henceforth omit means and standard deviations, reporting medians only.<sup>11</sup> The upper three rows in Table 3 contain the results for August 2011, and the lower three rows those for February 2012. The p-values of the Wilcoxon rank sum tests, reported for each metric and each month, show the probability that market makers have the same median as opportunistic traders.

To understand our measure of market-maker presence properly, consider one stock on one trading day. For each such stock-day, we calculate the average presence of those

---

<sup>10</sup> The 20% cut-off point is arbitrary. We repeat the analysis below with a 40% cut-off point, and obtain qualitatively similar results. Furthermore, we repeat the latter analysis with opportunistic HFTs defined as members with a less than 5% market-maker presence on average. Thus, potential borderline cases with a market-maker presence of between 5% and 40% are excluded. This setting changes none of the qualitative results, but the differences between market-maker HFTs and opportunistic HFTs are amplified. The results of the robustness tests are available from the authors upon request.

<sup>11</sup> In general, the medians of our statistics are close to the means, indicating symmetric distributions. For the latency and duration measures, however, the means often exceed the medians substantially. As this is likely to be due to large positive outliers in latency and duration, we report the medians only.

members classified as market makers and the average presence of those classed as opportunistic traders. For market makers, the median average presence across all such stock-days in August 2011 is 57%, and across those in February 2012 it is 59%. The low market-maker presence recorded for opportunistic HFTs, around 5% in both months, indicates that our subdivision of HFT trader types is indeed successful.

It may seem puzzling that opportunistic HFTs are on the supply-side of more than 30% of their trading volume, and yet supply liquidity in only 5% of the time. Note, however, that the trading volume of opportunistic HFTs is much lower than for market-making HFTs, and that it is spread out across more member firms. Furthermore, it may be the case that opportunistic HFTs utilize aggressive liquidity supply, i.e., limit orders that narrow the bid-ask spread. Such orders have high probabilities of execution, making them short-lived and less likely to be captured by our limit order book snapshots.

Brogaard (2011) measures the fraction of calendar time when *any* of the HFTs in his sample has quotes posted at the inside spread. Our measure, in contrast, is the market maker presence averaged across HFTs. For his sample of US stocks in 2010, Brogaard (2011) reports medians of 56%, 60%, and 94% for small-, medium-, and large-cap stocks respectively. Our evidence shows that a disaggregation of such numbers into individual members uncovers large diversity among HFTs.

Focusing on HFT activity, Table 3 shows that the lion's share of the HFT activity can be traced to market making. In August 2011, 71% of the HFTs' trading volume (SEK) and 81% of the limit order traffic was due to market-making activity. In February 2012, the share of HFTs' trading volume conducted by market makers was lower (65%), but the share of limit orders was higher (86%). To our knowledge, this is the first evidence of how HFT activity is distributed between market makers and opportunistic traders.

In accordance with the trading activity figures, the  $q/t$  ratios are significantly higher for the market makers than for the opportunistic traders. The difference is larger in February 2012 than in August 2011. In fact,  $q/t$  for the opportunistic HFTs (4.26) in February 2012 is closer to that for the non-HFTs (1.46) than to that for the market-making HFTs (10.78).

The *Liquidity supply* ratios show that market makers are on the passive side in 68% (71%) of their trades in August 2011 (February 2012), which is slightly less than the 74-79% reported by Jovanovic and Menkveld (2012). Furthermore, the median */Inventory/* levels, at 6% and 13% in the respective months, are in line with Jovanovic and Menkveld's (2012) characterization of HFT middlemen with strong inventory mean-reversion. According to their theoretical model, which they back up with empirical evidence, such middlemen have a positive impact on social welfare (under all reasonable parameter values).

The HFT described as a market maker by Menkveld (2013) engaged in cross-market activities between Chi-X and Euronext and tended to have a net inventory close to zero across those markets. Our */Inventory/* results indicate that the market-making HFT activity is almost in balance within NOMX-St, which is in line with a local market-making strategy. As we cannot observe inventory levels across trading venues, we do not know to what extent the market making HFTs engage in cross-market activities. The opportunistic HFTs have much higher values of */Inventory/*, 38% in August 2011 and 50% in February 2012, indicating that cross-market arbitrage activities are common in that group.

The minimum latency for the market makers is roughly the same in August 2011 and February 2012, at 0.1 milliseconds (ms). The opportunistic traders improve their

minimum latency between the two months, from 0.6 ms to 0.4 ms, but remain significantly slower than the market makers. The opportunistic HFTs' latency is more similar to the non-HFTs' latency (0.7 ms in August 2011 and 0.3 ms in February 2012) than to the market-making HFTs' latency. The low latency observed for the market-making HFTs serves as an illustration of the competition in liquidity supply in modern equity markets. Unless market makers respond immediately to news, they risk having their outstanding quotes picked off by quicker traders. The pick-off risk translates into adverse selection costs, forcing the market maker to charge wider (uncompetitive) spreads. For HFTs with strategies that rely on active rather than passive trading, latency may be less critical, which is reflected in our results.

The average limit order duration is longer in the less volatile month and longer for the market makers than for the opportunistic traders. This effect is likely to be related to the more passive strategy of market makers, and the fact that quotes require less frequent updating in a less volatile market environment.

Overall, the differences reported between the market makers and the opportunistic traders are substantial. For several metrics, the differences between the HFT groups are larger than the differences between the HFTs and non-HFTs reported in Table 2 (e.g., liquidity supply ratios in both months). This shows that the distinction between trader groups within the group of HFTs is just as important as the distinction between HFT and non-HFT activity.

### **4.3 HFT activity in different segments of stocks**

To further improve our understanding of HFT activity, we now split the thirty stocks into different market quality segments. The purpose of this analysis is to see whether the different types of HFTs choose the stocks in which they are active according to the

aggregate stock activity and stock market quality. Note, however, that the causality could be the reverse, i.e., that HFTs might affect the market quality of the stocks in which they are active.

For each sample month, we rank the OMXS 30 stocks according to their market capitalization, trading volume, realized volatility, and relative bid-ask spread in the previous month. For each ranking variable, we treat the upper and lower terciles as subsamples, and for each subsample we report the same metrics as in Table 3. The results are presented in Table 4, with market maker HFT activity in Panel A and opportunistic HFT activity in Panel B. The p-value of the z-test performed for each metric and each sorting indicates the probability that the upper and lower terciles of stocks have the same medians. For brevity, we only present results for February 2012.

The results in Panel A show that the market maker presence is significantly higher in stocks with higher market capitalization, higher trading volumes, lower realized volatility, and higher relative bid-ask spreads. These results are in line with market-making theory (see Madhavan, 2000, for an overview). Accordingly, market makers profit from high volumes and large spreads. Market making is more expensive in volatile stocks as inventory costs increase with risk, and in smaller stocks because they are less transparent, making private information more likely.

*Liquidity supply* only differs significantly across subsamples of stocks in the case of the bid-ask spread segments, where market makers supply less liquidity in those stocks where their market-maker presence is higher (Table 4, Panel A). Thus, increased market-maker presence does not imply an increased liquidity supply. That market makers should not be regarded as pure liquidity suppliers is in line with Kirilenko et al.

(2011). They argue that market makers demand liquidity either to offload unwanted inventory positions or to accumulate inventory under the expectation of a price change.

The market-making HFTs'  $q/t$  ratios, as well as their average limit order durations, are significantly lower in the segments where their market-maker presence is higher. These results may reflect a higher intensity of trading activity.

The results on opportunistic HFT activities in different segments of stocks are presented in Table 4, Panel B. The trading and limit order volumes mirror those of the market-making HFTs by definition, but for the liquidity supply ratios there is no mechanical relation to the market makers' results. Opportunistic HFTs supply significantly less liquidity for stocks with low market capitalization, low trading volume, high volatility, and large spreads. For the same stocks, the average limit order duration is significantly lower than in stocks with high market capitalization, high trading volume, low volatility, and tight spreads. For arbitrageurs, the main incentive for posting a limit order is the ability to offset a passive trade (when the limit order is hit by a market order) in one market with an active trade at a better price in another market (e.g., in another venue or in a related product). For OMXS 30 stocks, the active part of the strategy is likely to take place on the NOMX-St, as it tends to be more liquid than alternative trading venues for the same stocks and derivatives markets where related assets are traded. Hence, the higher opportunistic HFT liquidity demand seen in small, low-turnover, high-volatility, and high-spread stocks may be explained by higher arbitrage activity. It may, however, also reflect momentum ignition strategies, which are more likely to succeed in illiquid and volatile environments.

The analysis presented in this subsection sheds light on the behavior of HFTs in different segments of stocks. For policy making, a more important question is arguably what

causal influence HFT activities have on market quality. We address this question in the next section.

## **5 IMPACT OF HFT ON VOLATILITY: AN EVENT STUDY**

In this section, we study the influence of market-making HFT on volatility, which is probably the aspect of market quality that is most debated in relation to HFT activity (see Boehmer et al., 2012). An inherent problem when assessing the influence of HFT on volatility is the difficulty of determining the direction of causality. While it could be that HFT activity affects volatility, it is also well known that HFT strategies tend to focus on the most liquid stocks in each market, where efficiency is typically higher and volatility smaller than in less liquid stocks. We conduct an event study, using tick size changes as exogenous events, to disentangle the direction of causality. On the NOMX-St the minimum tick size depends on the price level. Specifically, the tick size for stocks priced below SEK 50 is SEK 0.01, that for stocks priced between SEK 50 and SEK 100 is five times higher, at SEK 0.05, and that for stocks priced between SEK 100 and SEK 500 is SEK 0.10. Whenever the stock price breaks through either the SEK 50 or SEK 100 price level, the minimum tick size changes.

There are two reasons to believe that tick size changes are useful in analyzing how HFTs impact volatility. Firstly, as HFTs have a relatively high turnover of their capital, and incur transaction costs relatively often, they are likely to be more sensitive to changes in bid-ask spread than non-HFTs (see, e.g., Chalmers and Kadlec, 1998). Secondly, it is reasonable to believe that tick size changes have different effects on market-making HFTs and opportunistic HFTs. To see why, consider a stock whose price moves from SEK 49 to SEK 51, causing the minimum bid-ask spread to increase from around 0.02% to around 0.1%. If the tick size at 0.1% is binding, then market making will be more

profitable, and liquidity-demanding strategies, such as arbitrage trading, become more costly. This reasoning is supported by Grossman and Miller (1988) and Harris (1997) in a pre-HFT market setting. Moreover, Goldstein and Kavajecz (2000) state that tick size changes have important implications for the supply of liquidity, trading by different market participants, and order submission strategies. Accordingly, we expect market-making (opportunistic) HFTs to increase (decrease) their share of trading activity following an increase in the minimum tick size.

Although tick size changes constitute exogenous shocks to HFT activity, previous studies show that they are also expected to change market quality and volatility directly (see, e.g., Goldstein and Kavajecz, 2000; Jones and Lipson, 2001). To overcome this issue, we divide the sample into two parts, before and after January 1, 2011 (henceforth Period I and Period II respectively). The motivation for this division is that, in 2010, the market-making HFTs identified in our sample had not yet entered the NOMX-St, or had negligible trading activity on it. Following the NOMX-St's efforts to reduce trading costs and latency in early 2010, including central counterparty clearing, capped fees, and the introduction of INET, the entry of market-making HFTs happened gradually in early 2011. Thus, during 2010, any HFT activity observed is due to opportunistic HFTs. After January 1, 2011, the HFT activity spans both market-making and opportunistic HFTs. We use Period I to quantify the combined volatility effect due to tick size changes and actions by opportunistic HFTs. We then use Period I as a control for Period II, making it possible to isolate the volatility effect of market-making HFTs. In other words, we regard the volatility effect in Period I as a joint effect of tick size and opportunistic HFTs, and the volatility effect in Period II as a joint effect of tick size, opportunistic HFTs, and

market-making HFTs. The difference between the two then constitutes a market-making HFT effect on volatility.

## 5.1 The events

We construct the events in the following manner. First, we identify an event day as a trading day on which a stock's transaction price either increases or decreases sufficiently to cross a tick size boundary, triggering a change in the minimum tick size. The tick size change takes effect at the moment when the price crosses the critical boundary, but we do not know how quickly different trader groups react to this. To ensure that all traders have time to react to the change, we identify the last day before and the first day after the event day on which transaction prices remain in the same tick size category for the whole day (we refer to these days as the *Before Day* and the *After Day*). If no such day is found within ten trading days before/after the potential event, the event is discarded. Moreover, the event is discarded if the *Before Day* and the *After Day* have stock prices in the same tick size category. That is, the event is only valid if it leads to a price that stays in the new tick size category for at least one full trading day within the next ten trading days.

To get enough events for this piece of analysis, we go beyond the two months studied above and use the full sample, from February 8, 2010 to March 31, 2012.<sup>12</sup> We identify 89 valid events during the sample period. Period I (before January 1, 2011) contains 42 events and Period II (after January 1, 2011) contains 47. The events occur in 20 out of the 30 stocks considered. The average number of trading days between the *Before Day* and the *After Day* is 2.25.

---

<sup>12</sup> There are no changes in the composition of the OMXS 30 index during the time period considered. The relation between tick size and price level referred to above is in accordance with the Federation of European Exchanges (FESE), Table 2, and is applied to OMXS 30 stocks from October 26, 2009 onward.

The aim of our event study is to compare the *Low-Tick* regime with the *High-Tick* regime. For events where the tick size is increasing, the Before day is taken to represent the Low-Tick regime, and the After day the High-Tick regime. When the tick size is decreasing, the Before day is treated as the High-Tick regime, and the After day as Low-Tick regime. We calculate all event statistics as changes from the Low-Tick regime to the High-Tick regime, regardless of the chronological order of the regimes. To check the accuracy of our expectations about how different HFTs adapt to changes in tick size, we first study changes in trading and quoting activity. After that, we turn to analyzing how volatility is affected by the tick size events.

## 5.2 Analysis of HFT event activity

Figure 2 shows the median changes in the trading activity of HFTs and non-HFTs corresponding to increases in the minimum tick size, for all events during the entire sample period. Evidently, both trader groups decrease their trading and quoting (limit order submissions and cancellations) activities in response to the increased tick size. In line with our expectations, and importantly for our event study, the HFTs' reactions are stronger than those of the non-HFTs, particularly in terms of trading volumes.

When evaluating the market quality impact of changes in trader group activities the change in market share is arguably a more relevant number than the change in nominal activity. For example, even though the trading volume of the non-HFTs falls when the tick size increases (see Figure 2), the market share of the non-HFTs actually increases, as the fall in the HFTs' trading volume is much sharper. That is, the weight of non-HFTs in the market increases, implying that their influence on market quality also increases. A focus on market share changes also allows for a straightforward comparison across stocks.

Denoting the stock trading volume corresponding to a trader group  $j$  on a day  $t$  as  $tvol_{j,t}$ , with  $j = [All\ HFTs, Opportunistic\ HFTs, Market-making\ HFTs]$ , we calculate trader  $j$ 's share of all three groups' aggregate daily trading volumes in the stock as:

$$(1) \quad \theta_{j,t}^{vol} = \frac{tvol_{j,t}}{tvol_t}$$

where  $tvol_t$  is the total trading volume for non-HFTs and HFTs combined.

To observe the effect of a tick size change for an event stock, we calculate  $\theta_{j,t}^{vol}$  for the High-Tick and Low-Tick regime days for each event ( $t = [High, Low]$ ). In order to control for potential time-series variations in HFT activities in the event stock that are unrelated to the tick size event, we also obtain the averages of the corresponding differences for all other OMXS 30 (*benchmark*) stocks on the same days.<sup>13</sup> Moreover, to further highlight how trading activity reactions to tick size events differ across trader groups, we define a market share change as  $\tilde{\theta}_j^{vol} = \theta_{j,High}^{vol} / \theta_{j,Low}^{vol}$ , for both the event stock and the benchmark stocks. Finally, to get a metric of abnormal changes in trading activity related to the tick size event, we calculate the difference in the market share changes, between the event stock and the benchmark stocks. Accordingly, our metric for an abnormal change in market share of trading volume is:

$$(2) \quad \Delta \tilde{\theta}_j^{vol} = [\tilde{\theta}_j^{vol}]_{Event} - [\tilde{\theta}_j^{vol}]_{Benchmark}$$

As difference-in-difference metrics may be difficult to interpret, we provide a breakdown of the abnormal trading volume market share in Table 5. In the first row, we report the market shares used to calculate the Period I abnormal change in the trading volume of all HFTs. For the event stocks, the HFT median market share of trading

---

<sup>13</sup> Stocks that are involved in another qualified event with Before and After Days that to any extent overlap with the current event's Before and After Days are excluded from the control group.

volume is 0.132 in the High-Tick regime and 0.246 in the Low-Tick regime, resulting in a median High/Low ratio  $\tilde{\theta}_{All\ HFTs, Event}^{tvol} = 0.567$ . This indicates that HFTs have a larger market share of the trading volume when the minimum tick size is smaller. For the benchmark stocks, we observe a very small difference between the HFT median share of trading volume in the High-Tick regime (0.177) and that in the Low-Tick regime (0.188). Consequently, the median High/Low ratio is very close to 1 ( $\tilde{\theta}_{All\ HFTs, Benchmark}^{tvol} = 0.987$ ). Comparing the event stocks with the benchmark, we obtain the HFT median abnormal trading volume market share,  $\Delta\tilde{\theta}_{All\ HFTs}^{tvol} = -0.435$ .

Throughout our event study, we make the abnormal change our primary metric, reflecting a difference-in-difference approach. For example,  $\Delta\tilde{\theta}_{All\ HFTs}^{tvol}$  is the abnormal HFT market share effect for a stock that experiences a tick size change event. It reflects the High-Tick regime market share *relative* to the Low-Tick regime market share, *relative* to the average corresponding fraction of these two respective market shares across the set of benchmark stocks.

In Period II (see Table 5, 2<sup>nd</sup> row), the HFTs have a Low-Tick median market share equal to 0.416, and a corresponding High-Tick median market share of 0.361, resulting in a median High/Low ratio of 0.889. For the benchmark stocks, the median High/Low ratio is equal to 0.978, and the median abnormal volume market share is -0.036. That is, as a group, HFTs appear to react less strongly to tick size changes in Period II than in Period I. However, recalling the discussion above, Period I and Period II are associated with different sets of HFTs on the NOMX-St. Whereas all of the HFTs that are active during Period I are categorized as opportunistic, the Period II HFTs also include market makers. In line with our expectations, a disaggregation of the HFT results shows that the two subcategories tend to react in opposite ways to tick size changes. The opportunistic

HFTs have almost identical median High/Low volume market share ratios and abnormal trading volume market share changes in Periods I and II. The market-making HFTs, on the other hand, have a higher median share of the trading volume in the High-Tick regime than they do in the Low-Tick regime. Thus, their median High/Low ratio is larger than unity (1.169), and the median abnormal volume market share change is positive (0.161). These results are in line with our predictions on the relative profitability of strategies in different tick size regimes. The strongly negative effect on the opportunistic HFTs' market share is due to the fact that liquidity-demanding strategies are less profitable when the tick size is larger. Market-making profitability, on the other hand, is boosted by a higher minimum tick size, which is reflected in increased market shares for market-making HFTs. The results on the market-making HFT activity are also consistent with our empirical finding that the market-making HFTs have a higher market maker presence when it comes to stocks with large bid-ask spreads.

In Table 6, we present the statistical tests that we use to determine whether the abnormal changes in market share are significant. The left-hand column contains the abnormal changes as calculated in Table 5, along with p-values (in parentheses) from the Wilcoxon signed rank test. The tests indicate that the Period I HFT effect is statistically different from zero. In Period II, the aggregate HFT effect is not significantly different from zero. When analyzed separately, however, both the opportunistic and the market-making HFT effects are statistically significant.

In the next two columns of Table 6, we split the trading volume results into active and passive trading volumes. Trading volume is considered to be active when the trader in question initiates the trade by submitting a marketable order. A trade that occurs when the limit order of the trader in question is hit by an incoming order is considered

passive. The abnormal market share changes for these variables are calculated in the same way as for total trading volume. Notably, in Periods I and II alike, the decrease in the trading activity of the opportunistic HFTs can be traced to a decrease in active trading – their market share of liquidity supply volumes do not decrease abnormally. This is in line with the notion that liquidity-demanding strategies are more expensive in High-Tick regimes. Moreover, we find that the increase in market-making HFTs' activity is due to both active and passive trading. The strong increase in the market-making HFTs' active trading volume market share may appear surprising, but it is likely to be due more to the sharp decrease in the trading of opportunistic HFTs than an increase in that of market-making HFTs.

In all the results above, the trading volumes are in terms of the SEK amount. As the size of trades may differ across trader groups, we repeat the analysis using trading volumes in terms of number of executions. The results, presented in Table 6, are qualitatively the same as those discussed above.

Overall, the results lend support to the use of tick size changes as exogenous events for HFT activity. Firstly, the reaction of HFTs to tick size events is stronger than the reaction of non-HFTs. Secondly, the tick size change produces an exogenous shock in the relative suitability of different HFT strategies, which can be seen in the adjustments to liquidity demand and supply for different HFT sub-groups. An increased minimum tick size makes strategies based on liquidity supply (such as market making) more attractive, whereas strategies that consume liquidity (such as arbitrage strategies) become more expensive. In the next section, we utilize the tick size events to investigate the causal influence of HFT activity on volatility.

### 5.3 Volatility analysis

The pros and cons of HFT activity in financial markets are widely debated. In the academic literature, most empirical evidence points to beneficial effects on liquidity and price discovery. For volatility, however, the evidence is mixed. In a study of NASDAQ stocks, Hasbrouck and Saar (2013) find that HFTs reduce short-term volatility. However, in a study of the flash crash of May 2010, when a large sell order in the S&P 500 E-mini futures market caused prices to temporarily fall by 9% in a few minutes, Kirilenko et al. (2011) find that HFTs may have amplified the crash. Furthermore, using a large international data set, Boehmer et al. (2012) find that short-term volatility is amplified by AT. Given this divergence in results, we focus our assessment of the effects of HFT activity on the effects on short-term volatility.

We calculate intraday realized stock return volatility, using midpoint quote changes at 1, 5, 10, and 15-minute intervals.<sup>14</sup> Let  $\sigma(k)_t$  denote the average intraday realized volatility, for  $k = 1, 5, 10$ , and 15 minutes, for trading day  $t$ . To observe the volatility effect of a tick size change we calculate  $\sigma(k)_t$  for the High-Tick and Low-Tick regimes, for the event stock as well as the benchmark stocks. We then apply exactly the same difference-in-difference approach as above. We define an abnormal volatility change as the ratio of the volatility of a stock that experiences a tick size change event in the High-Tick regime to the corresponding volatility under the Low-Tick regime, *relative* to the corresponding volatility ratio of the benchmark stocks:

$$(3) \quad \Delta\tilde{\sigma}(k) = \left[ \frac{\sigma(k)_{High}}{\sigma(k)_{Low}} \right]_{Event} - \left[ \frac{\sigma(k)_{High}}{\sigma(k)_{Low}} \right]_{Benchmark}.$$

---

<sup>14</sup> Harris (1990) shows that stock price discreteness might affect volatility. Accordingly, in our event study setting, a tick size change might induce a mechanical effect on volatility measured using changes in transaction prices. For example, an increase in the minimum tick size will increase the minimum bid-ask spread, which in turn might increase volatility through a larger amplitude in the bid-ask bounce between transactions. We use changes in bid-ask spread midpoint quotes in order to mitigate potential effects from price discreteness on volatility in the event study.

In order to isolate the impact of market-making HFT activity on volatility, we use Period I as a control for our analysis of Period II, and investigate whether  $\Delta\tilde{\sigma}(k)$  is, on average, different in the two periods. Thus, strictly speaking, we are able to disentangle the causal effect of market-making HFTs' activity on short-term volatility, while we are not able to say anything about the causality between the opportunistic HFTs' activity and volatility.

Table 7, Panel A, presents the results of the volatility analysis, along with statistical tests of the same type as in Table 6. For Period I, we note a positive abnormal volatility associated with the tick size increase. This effect is significantly different from zero at the 1% level for each of the four different realized volatility measures. In connection with the results from Table 6, our interpretation of this result is that an increase in the minimum tick size leads to a joint decrease in the opportunistic HFTs' trading activity and increase in stock return volatility.

For Period II, we observed in Table 6 that the opportunistic HFTs behaved in the same way as they did in Period I, but that the market-making HFTs increased their activities when the tick size increased. In Table 7, across all volatility measures, we find that the Period II abnormal volatility is not significantly different from zero. Thus, we conclude that a tick size increase during Period II is associated with a decrease in opportunistic HFTs' activity, an increase in market-making HFTs' activity, and an unchanged volatility.

Finally, in our effort to isolate the causal effect of market-making HFTs' activity on volatility, we perform a test of the null hypothesis that the median abnormal volatility effect is the same before and after January 1, 2011. From the results in Table 7, we note that the difference between the medians of  $\Delta\tilde{\sigma}(k)$  for Period I and Period II is significantly negative at the 5% (10%) level for the 5- and 10- (1- and 15-) minute volatility measures. Thus, the null hypothesis can be rejected, implying that the increase

in market-making HFTs' activity causes a decrease in short-term volatility. This is the key conclusion of our event study.

Several studies document an inverse relationship between stock prices and volatility, the first being Black (1976), which is commonly referred to as the *leverage effect*. Accordingly, volatility is likely to increase (decrease) when stock prices fall (rise). To control for a possible effect on volatility from stock price changes per se, rather than through altered HFT behavior or a tick size event, we employ an alternative method of obtaining the abnormal volatility in the events. Instead of using all non-event stocks to obtain the benchmark volatility ratio in Equation (3), we identify one non-event stock with an event return that matches that of the event stock, and obtain the volatility ratio for that stock alone. Specifically, we calculate the event return as the relative price change from the closing of the Before Day to the opening of the After Day. The non-event stock that has the event return closest to the event return of the event stock is used as the benchmark in this robustness test. Thus, the benchmark stock is exposed to a similar price change in the event, while only the event stock's price change triggers a tick size change.

The results of our robustness test with respect to the leverage effect are presented in Table 7, Panel B. Overall, the results from Panel A are robust to the change of benchmark. The abnormal volatility is significantly positive in the first part of the sample, and not significantly different from zero in the second part of the sample. In fact, the significantly positive median abnormal volatility in the first part of the sample is larger in Panel B than the corresponding figures in Panel A when volatility is measured at the one and five minutes frequencies. The null hypothesis that the median abnormal volatility during the first part of the sample is the same as the median abnormal

volatility during the second part of the sample can be rejected at the 5% (10%) level for the 1- and 5- (10-) minute volatility measures. However, when realized volatility is measured using 15-minute returns, the corresponding difference between medians has a negative sign, but it is not significantly different from zero. Overall, however, the robustness analysis with respect to the leverage effect confirms the result that increased market-making HFTs' activity has a soothing effect on short-term volatility.

## 6 CONCLUSION

This paper contributes to the HFT literature by subcategorizing HFT strategies into market-making HFT and opportunistic HFT, by providing a detailed analysis of their trading characteristics, and by assessing their influence on volatility. We find that a majority of the HFTs' trading volume and more than 80% of HFT limit order submissions on the NOMX-St are associated with market-making strategies. Market-making HFTs thus constitute an important group of traders in modern equity markets, which investors and regulators need to understand when choosing their policies. We find that market-making HFTs have higher order-to-trade ratios and lower latency than opportunistic HFTs. We think this is due to market makers continuously monitoring order flows and responding to news faster than other traders, thereby reducing their exposure to adverse selection (as argued by Jovanovic and Menkveld, 2012). Interestingly, opportunistic HFTs, including those using arbitrage and directional strategies, have order-to-trade ratios and latency on a par with non-HFTs.

Using an event study based on changes in minimum tick size, we find that market-making HFT activities are good for the overall market quality in the sense that they reduce short-term volatility. This finding is important, as previous empirical research has shown mixed conclusions on the AT and HFT impacts on volatility (whereas most

agree that the activity is beneficial for liquidity and price discovery). In practice, volatility affects both risk management policies and liquidity supply decisions among traders. Given our rich data set and exogenous event setting based on tick size changes, our evidence brings deeper insight to the policy debate on HFT.

Finally, our event study shows that a higher tick size decreases both the trading and quoting activities of all trader groups, in particular HFTs. An increased minimum tick size makes strategies based on liquidity supply (such as market making) more attractive, whereas strategies that consume liquidity (such as arbitrage strategies) become more expensive. Thus, tick size regulation may be an interesting solution for limiting quoting traffic without threatening the advantages of HFT activity. We consider this an interesting direction for future research.

## **ACKNOWLEDGEMENTS**

We would like to thank NASDAQ OMX for providing the data, and Petter Dahlström, Frank Hatheway and Björn Hertzberg for numerous discussions about the data and the market structure. We are also grateful to Anders Anderson, Richard Anderson, Mattias Hammarquist, Terrence Hendershott, Kjell Jørgensen, Pete Kyle, and Jonathan Tse for useful comments and discussions. Any remaining errors are our own. Both authors are grateful to the Jan Wallander and Tom Hedelius foundation and the Tore Browaldh foundation for research support.

## **REFERENCES**

Bessembinder, H., Panayides, M., Venkataraman, K., 2009. Hidden liquidity: an analysis of order exposure strategies in electronic stock markets. *J. Financ. Econ.* 94(3), 361–383.

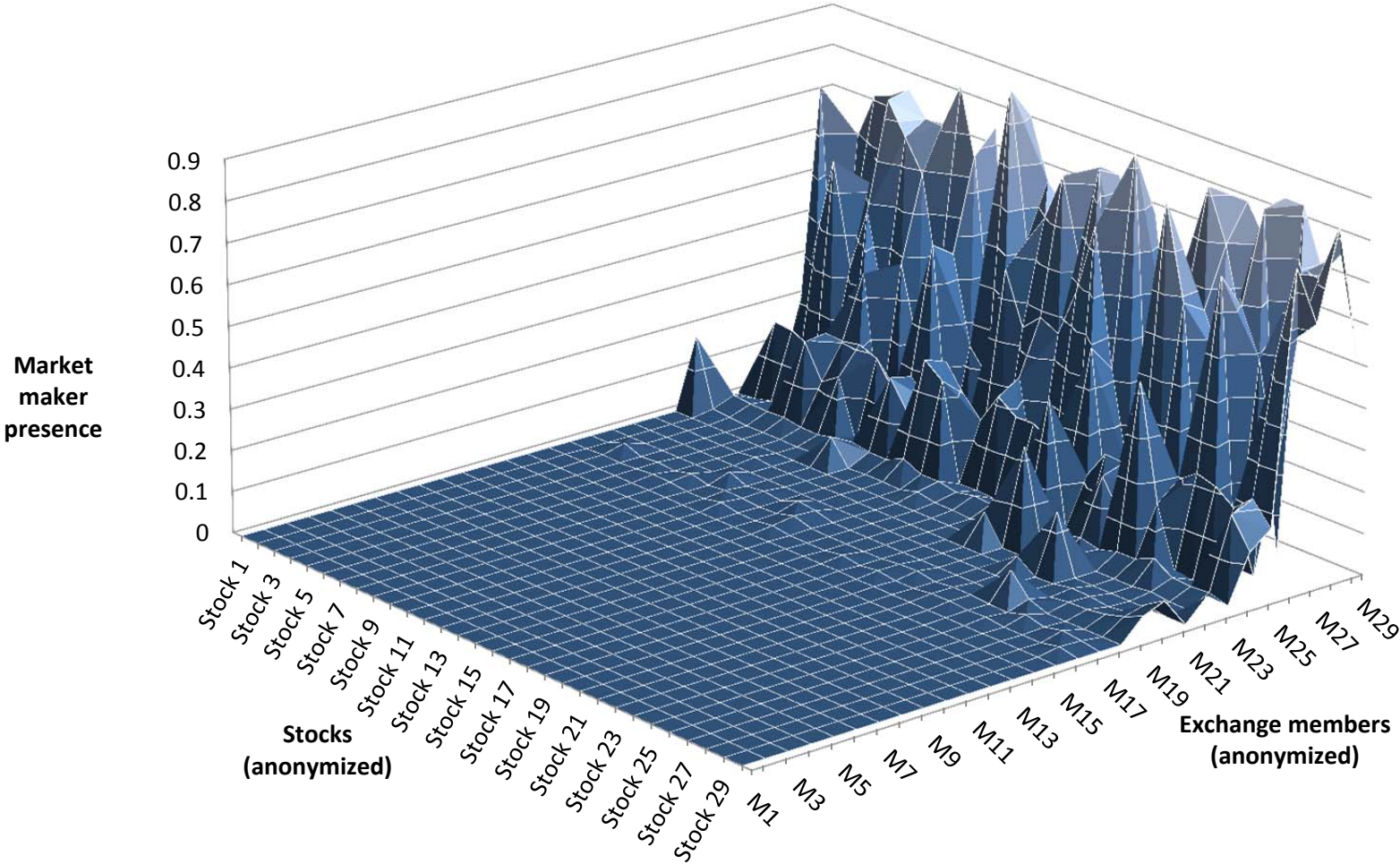
- Biais, B., Foucault, T., Moinas, S., 2011. Equilibrium high frequency trading. Working paper.
- Black, F., 1976. Studies of stock price volatility changes. *Proceedings of the Business and Economics Section of the American Statistical Association*, 177–181.
- Boehmer, E., Fong, K., Wu, J., 2012. International evidence on algorithmic trading. Working paper.
- Brogaard, J., 2011. The activity of high frequency traders. Working paper.
- Brogaard, J., Hendershott, T., Riordan, R., 2012. High frequency trading and price discovery. Working paper.
- Carrion, A., 2013. Very fast money: high-frequency trading on the NASDAQ. Working paper.
- Chalmers, J.M.R., Kadlec, G.B., 1998. An empirical examination of the amortized spread. *J. Financ. Econ.* 48, 159–188.
- Finansinspektionen. 2012. Investigation into high frequency and algorithmic trading. FI Report, February 2012. Available at [www.fi.se](http://www.fi.se) (retrieved on August 14, 2012).
- Goldstein, M., Kavajecz, K., 2000. Eighths, sixteenths, and market depth: changes in tick size and liquidity provision on the NYSE. *J. Financ. Econ.* 56, 125–149.
- Gomber, P., Arndt, B., Lutat, M., Uhle, T., 2011. High-frequency trading. Working paper.
- Grossman, S., Miller, M., 1988. Liquidity and market structure. *J. Financ.* 43, 617–633.

- Harris, L., 1990. Estimation of stock price variances and serial covariances from discrete observations. *J. Financ. Quant. Anal.* 25, 291–306.
- Harris, L., 1997. Decimalization: a review of the arguments and evidence. Working paper.
- Hasbrouck, J., Saar, G., 2013. Low-latency trading. Working paper.
- Hautsch, N., Huang, R., 2012. On the dark side of the market: identifying and analyzing hidden order placements. Working paper.
- Hendershott, T., Jones, C.M., Menkveld, A.J., 2011. Does algorithmic trading improve liquidity? *J. Financ.* 66(1), 1–33.
- Hendershott, T., Riordan, R., 2012. Algorithmic trading and the market for liquidity. *J. Financ. Quant. Anal.* Forthcoming.
- Hirschey, N., 2011. Do high-frequency traders anticipate buying and selling pressure? Working paper.
- Jarrow, R., Protter, P., 2011. A dysfunctional role of high frequency trading in electronic markets. Working paper.
- Jones, C., Lipson, M., 2001. Sixteenths: direct evidence on institutional execution costs. *J. Financ. Econ.* 59, 253–278.
- Jovanovic, B., Menkveld, A.J., 2012. Middlemen in limit order markets. Working paper.
- Kirilenko, A.A., Kyle, A.S., Samadi, M., Tuzun, T., 2011. The flash crash: the impact of high frequency trading on an electronic market. Working paper.

- Madhavan, A., 2000. Market microstructure: a survey. *J. Financ. Mark.* 3, 205–258.
- McInish, T., Upson, J., 2011. Strategic liquidity supply in a market with fast and slow traders. Working paper.
- Menkveld, A.J., 2013. High frequency trading and the new-market makers. Working paper.
- SEC. 2010. Concept release on equity market structure. Available at [www.sec.gov/rules/concept/2010/34-61358.pdf](http://www.sec.gov/rules/concept/2010/34-61358.pdf) (retrieved on September 25, 2012).

Figure 1: Market maker presence across OMXS 30 stocks and exchange members

Panel A: August 2011



Panel B: February 2012

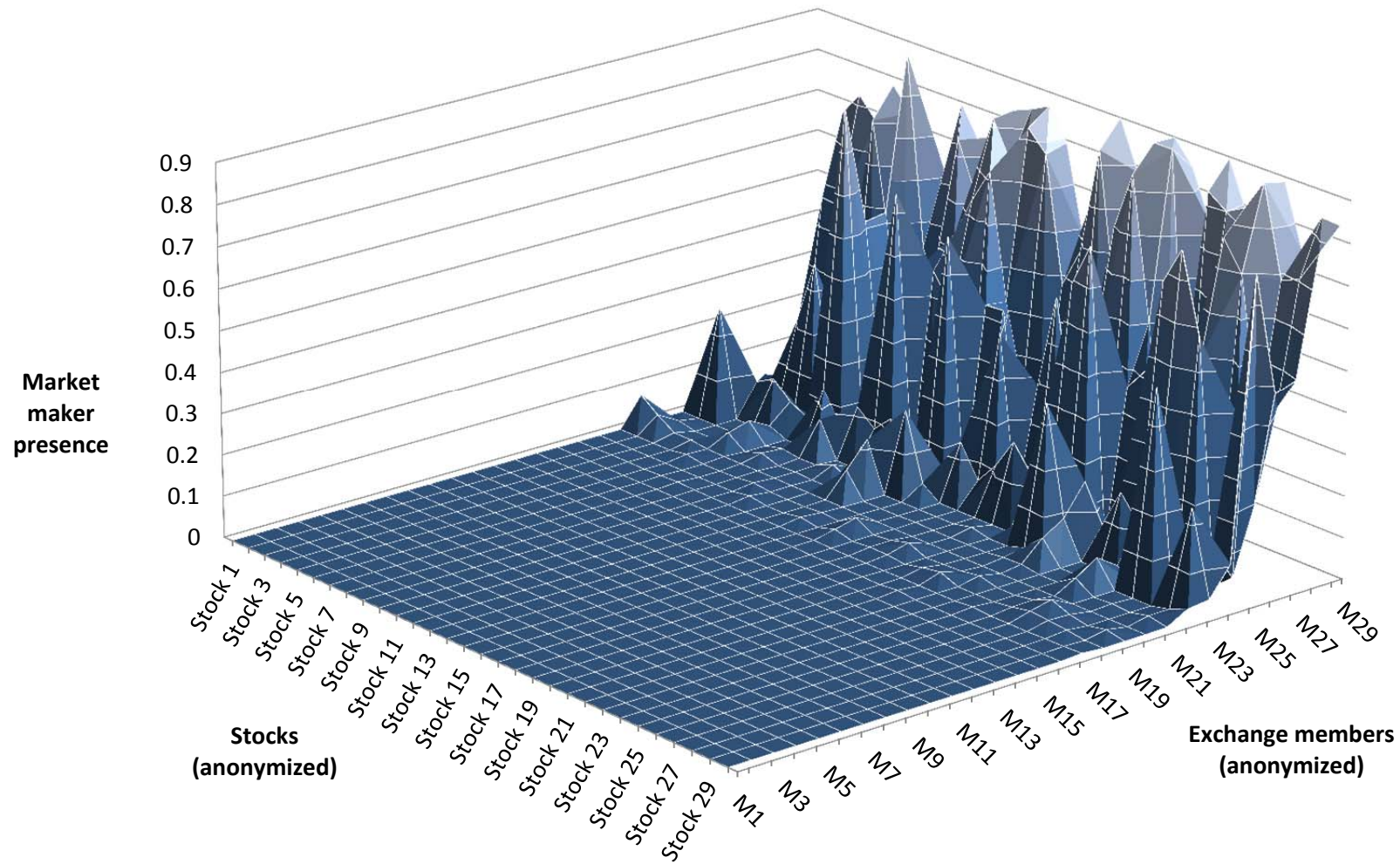


Figure 1 presents the market-maker presence for each exchange member in each OMXS 30 stock. Market-maker presence is estimated as the fraction of 10-second periods in the continuous trading day during which a member has quotes at either of the best bid and offer prices, reported as the mean across all trading days in August 2011 (Panel A) and February 2012 (Panel B). On the z-axis, exchange member identities (MPIDs) are sorted by their average market-maker presence across stocks each month.

Figure 2: Effect of increasing tick size on trading activity

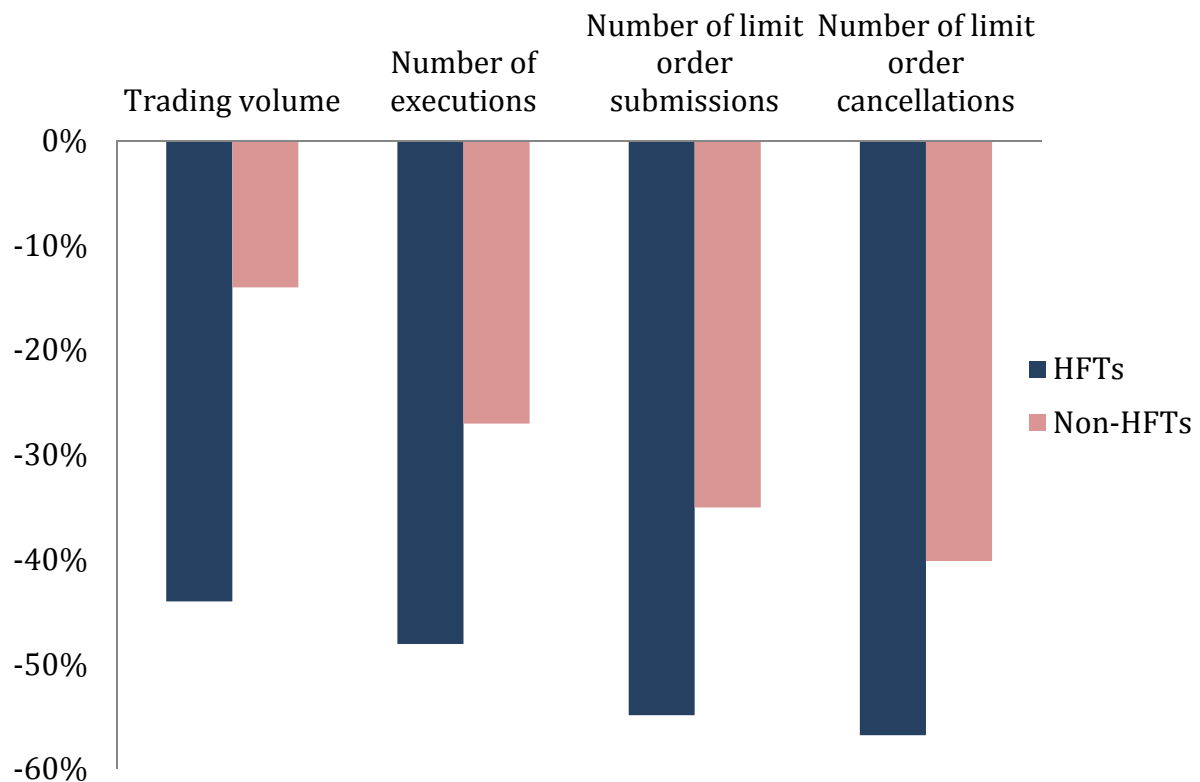


Figure 2 present changes in the trading activity of HFTs and Non-HFTs in stocks that experience tick size changes. For each tick size change, trading activity on the closest trading day when the lower tick size holds throughout the day (Low), and the closest trading day when the high tick size holds throughout the day (High), are compared. The bar chart shows the median activity change observed for an increasing minimum tick size in terms of SEK trading volume, number of executions, number of limit order submissions, and number of limit order cancellations. Reported statistics are based on 89 tick size events in the OMXS30 stocks during February 8, 2010 – March 31, 2012.

Table 1: Summary statistics

|          |        | <i>Market cap.</i><br>(MSEK) | <i>Trading volume</i><br>(MSEK) | <i>Number of trades</i> | <i>Number of traded shares</i><br>(x1000) | <i>Number of limit order submissions</i> | <i>Number of limit order cancellations</i> | <i>Depth</i> | <i>Nominal bid-ask spread</i><br>(SEK/100) | <i>Relative bid-ask spread</i><br>(%) | <i>Volatility</i> |
|----------|--------|------------------------------|---------------------------------|-------------------------|-------------------------------------------|------------------------------------------|--------------------------------------------|--------------|--------------------------------------------|---------------------------------------|-------------------|
| Aug 2011 | Mean   | 79,340                       | 527                             | 7,522                   | 4,936                                     | 261,478                                  | 302,826                                    | 9,016        | 12.39                                      | 0.104                                 | 0.215             |
|          | Median | 54,426                       | 396                             | 5,856                   | 2,857                                     | 243,340                                  | 260,491                                    | 7,951        | 13.41                                      | 0.099                                 | 0.223             |
|          | St Dev | 69,434                       | 398                             | 4,409                   | 4,657                                     | 141,834                                  | 177,248                                    | 4,810        | 7.93                                       | 0.025                                 | 0.078             |
| Feb 2012 | Mean   | 91,372                       | 361                             | 4,962                   | 2,985                                     | 82,036                                   | 97,398                                     | 14,247       | 11.97                                      | 0.086                                 | 0.044             |
|          | Median | 63,476                       | 298                             | 4,347                   | 1,967                                     | 68,823                                   | 63,668                                     | 13,469       | 12.96                                      | 0.091                                 | 0.042             |
|          | St Dev | 77,330                       | 210                             | 2,122                   | 2,662                                     | 58,412                                   | 97,599                                     | 6,589        | 5.82                                       | 0.021                                 | 0.021             |
|          | t-test | 0.5285                       | 0.0493                          | 0.0058                  | 0.0511                                    | 0.0000                                   | 0.0000                                     | 0.0009       | 0.8148                                     | 0.0049                                | 0.0000            |
|          | z-test | 0.3871                       | 0.0993                          | 0.0073                  | 0.0215                                    | 0.0000                                   | 0.0000                                     | 0.0014       | 0.9941                                     | 0.0127                                | 0.0000            |
|          | F-test | 0.5657                       | 0.0010                          | 0.0002                  | 0.0036                                    | 0.0000                                   | 0.0019                                     | 0.0955       | 0.1017                                     | 0.2615                                | 0.0000            |
|          |        |                              |                                 |                         |                                           |                                          |                                            |              |                                            |                                       |                   |

Table 1 presents summary statistics for the sample of OMXS 30 index stocks during August 2011 and February 2012. Each mean, median, and standard deviation is an equally weighted estimate across stocks. For each month and each stock; *Market cap* is the number of outstanding shares times the last transaction price of the month (in SEK 1,000,000); *Trading volume* is the total value of shares traded per day (in SEK 1,000,000); *Number of trades* is the number of transactions per day during the continuous trading session; *Number of traded shares* is the number of traded shares per day ( $\times 1,000$ ) during the continuous trading session; *Number of limit order submissions* is the number of orders posted to the limit order book per day during the continuous trading session; *Number of cancellations* is the number of cancelled orders (including partial cancellations) from the limit order book per day during the continuous trading session; *Depth* is defined as the number of shares needed to be bought or sold in order to move the ask or bid price by 1%, reported as the average of ask and bid depth using end-of-minute snapshots of the order book for each stock; *Nominal bid-ask spread* is the average best quoted bid-ask spread using end-of-minute snapshots of the order book for each stock (in SEK 0.01); *Relative bid-ask spread* is the average best quoted bid-ask spread, divided by the spread midpoint, using end-of-minute snapshots of the order book for each stock (%); *Volatility* is the intraday realized stock return volatility, using midpoint quote changes in one-minute intervals. The last three rows contain p-values from a t-test for equality between means, a Wilcoxon rank sum z-test for equality between medians and an F-test for equality between variances across the two months. Statistics for August 2011 and February 2012 are based on 23 and 21 trading days, respectively.

Table 2: HFT vs. non-HFT characteristics

Panel A: August 2011

|         |        | <i>Fraction of all<br/>trading<br/>volume</i> | <i>Fraction<br/>of all add<br/>orders</i> | <i> Inventory </i> | <i>q / t</i> | <i>Minimum<br/>latency<br/>(ms)</i> | <i>Limit order<br/>duration<br/>(ms)</i> | <i>Liquidity<br/>supply<br/>(volume)</i> | <i>Liquidity<br/>supply<br/>(trades)</i> |
|---------|--------|-----------------------------------------------|-------------------------------------------|--------------------|--------------|-------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| HFT     | Mean   | 0.29                                          | 0.30                                      | 0.27               | 13.82        | 0.36                                | 2774                                     | 0.59                                     | 0.58                                     |
|         | Median | 0.29                                          | 0.29                                      | 0.27               | 12.91        | 0.38                                | 3031                                     | 0.59                                     | 0.58                                     |
|         | St Dev | 0.01                                          | 0.04                                      | 0.03               | 2.16         | 0.03                                | 1149                                     | 0.02                                     | 0.02                                     |
| Non-HFT | Mean   | 0.31                                          | 0.05                                      | 0.58               | 2.68         | 0.96                                | 20434                                    | 0.56                                     | 0.49                                     |
|         | Median | 0.30                                          | 0.05                                      | 0.59               | 2.52         | 0.74                                | 19197                                    | 0.56                                     | 0.50                                     |
|         | St Dev | 0.04                                          | 0.01                                      | 0.03               | 0.51         | 0.55                                | 7518                                     | 0.03                                     | 0.02                                     |
|         | t-test | 0.008                                         | 0.000                                     | 0.000              | 0.000        | 0.000                               | 0.000                                    | 0.000                                    | 0.000                                    |
|         | z-test | 0.007                                         | 0.000                                     | 0.000              | 0.000        | 0.000                               | 0.000                                    | 0.000                                    | 0.000                                    |
|         | F-test | 0.000                                         | 0.000                                     | 0.924              | 0.000        | 0.000                               | 0.000                                    | 0.098                                    | 0.459                                    |

Panel B: February 2012

|         |        | <i>Fraction of<br/>total trading<br/>volume</i> | <i>Fraction of<br/>all order<br/>submissions</i> | <i> Inventory </i> | <i>q / t</i> | <i>Minimum<br/>latency<br/>(ms)</i> | <i>Limit order<br/>duration<br/>(ms)</i> | <i>Liquidity<br/>supply<br/>(volume)</i> | <i>Liquidity<br/>supply<br/>(trades)</i> |
|---------|--------|-------------------------------------------------|--------------------------------------------------|--------------------|--------------|-------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| HFT     | Mean   | 0.25                                            | 0.30                                             | 0.35               | 8.94         | 0.21                                | 7627                                     | 0.57                                     | 0.58                                     |
|         | Median | 0.26                                            | 0.30                                             | 0.36               | 8.97         | 0.21                                | 7273                                     | 0.57                                     | 0.58                                     |
|         | St Dev | 0.03                                            | 0.06                                             | 0.03               | 0.72         | 0.02                                | 2151                                     | 0.02                                     | 0.03                                     |
| Non-HFT | Mean   | 0.35                                            | 0.06                                             | 0.62               | 1.51         | 0.49                                | 33788                                    | 0.50                                     | 0.51                                     |
|         | Median | 0.36                                            | 0.06                                             | 0.62               | 1.46         | 0.32                                | 32856                                    | 0.50                                     | 0.51                                     |
|         | St Dev | 0.03                                            | 0.01                                             | 0.02               | 0.21         | 0.49                                | 6283                                     | 0.02                                     | 0.02                                     |
|         | t-test | 0.000                                           | 0.000                                            | 0.000              | 0.000        | 0.018                               | 0.000                                    | 0.000                                    | 0.000                                    |
|         | z-test | 0.000                                           | 0.000                                            | 0.000              | 0.000        | 0.000                               | 0.000                                    | 0.000                                    | 0.000                                    |
|         | F-test | 0.894                                           | 0.000                                            | 0.001              | 0.000        | 0.000                               | 0.000                                    | 0.783                                    | 0.161                                    |

Table 2 presents summary statistics of the trading and quoting activity of HFTs and non-HFTs members in OMXS 30 index stocks during August 2011 (Panel A) and February 2012 (Panel B). All statistics are reported for time series of daily frequency, where the daily observations are calculated as follows. The *Fraction of all trading volume* is the sum of SEK volume traded for all members in each group (HFT and non-HFT) in all stocks, divided by the sum of total SEK trading volume in all stocks; the *Fraction of all order submissions* is the sum of order submissions posted for all stocks by members in each group, divided by the total number of order submissions orders in all stocks; *|Inventory|* is the absolute accumulated inventory divided by the trading volume (in number of shares), reported as the median across members in each group and all stocks; *q/t* is the aggregate number of add orders in all stocks divided by the aggregate number of executions in all stocks, for each group; *Minimum latency (Limit order duration)* is the minimum (median) lifetime of limit orders that are canceled before the end of the trading day, reported in milliseconds and as the median across all stocks as well as members of each trader group; *Liquidity supply* is the fraction of the group's trading volume in all stocks where the member is on the passive side, counted in either SEK trading volume or number of trades. The last three rows contain p-values from a t-test for equality between means, a Wilcoxon rank sum z-test for equality between medians and an F-test for equality between variances across the two groups. As there are 23 (21) trading days in August 2011 (February 2012), and 30 stocks, the number of observations in each statistical test is 690 (630).

Table 3: Market-making vs. opportunistic HFTs

|                                | <i>Market-maker presence</i> | <i>Fraction of HFT trading volume</i> | <i>Fraction of HFT order submissions</i> | <i> Inventory </i> | <i>q / t</i> | <i>Minimum latency (ms)</i> | <i>Limit order duration (ms)</i> | <i>Liquidity supply (volume)</i> | <i>Liquidity supply (trades)</i> |
|--------------------------------|------------------------------|---------------------------------------|------------------------------------------|--------------------|--------------|-----------------------------|----------------------------------|----------------------------------|----------------------------------|
| <i>August 2011 (medians)</i>   |                              |                                       |                                          |                    |              |                             |                                  |                                  |                                  |
| Market-making HFTs             | 0.57                         | 0.71                                  | 0.81                                     | 0.06               | 14.34        | 0.10                        | 3520                             | 0.68                             | 0.66                             |
| Opportunistic HFTs             | 0.05                         | 0.29                                  | 0.19                                     | 0.38               | 10.26        | 0.60                        | 2687                             | 0.36                             | 0.32                             |
| z-test (p-value)               | 0.000                        | 0.000                                 | 0.000                                    | 0.000              | 0.016        | 0.000                       | 0.020                            | 0.000                            | 0.000                            |
| <i>February 2012 (medians)</i> |                              |                                       |                                          |                    |              |                             |                                  |                                  |                                  |
| Market-making HFTs             | 0.59                         | 0.65                                  | 0.86                                     | 0.13               | 10.78        | 0.09                        | 10754                            | 0.71                             | 0.70                             |
| Opportunistic HFTs             | 0.05                         | 0.35                                  | 0.14                                     | 0.50               | 4.26         | 0.36                        | 3252                             | 0.32                             | 0.30                             |
| z-test (p-value)               | 0.000                        | 0.000                                 | 0.000                                    | 0.000              | 0.000        | 0.000                       | 0.000                            | 0.000                            | 0.000                            |

Table 3 presents the medians of the trading and quoting activity of market makers and opportunistic traders within the group of HFT members, for OMXS 30 index stocks, during August 2011 and February 2012. Medians are reported for time series of daily frequency, where the daily observations are calculated as follows. *Market-maker presence* is the fraction of 10-second periods in the continuous trading session during which a member has quotes at either of the best bid and offer prices, averaged across members in each group (market-making HFTs and opportunistic HFTs) and all stocks; the *Fraction of HFT trading volume* is the sum of SEK volume traded for all members in each group in all stocks, divided by the total HFT SEK trading volume in all stocks; the *Fraction of HFT order submissions* is the sum of order submissions posted for all stocks by members in each group, divided by the total number of HFT order submissions in all stocks; *|Inventory|* is the absolute accumulated inventory divided by the trading volume (in number of shares), reported as the median across members in each group and across all stocks; *q/t* is the aggregate number of add orders in all stocks divided by the aggregate number of executions in all stocks, for each group; *Minimum latency (Limit order duration)* is the minimum (median) lifetime of limit orders that are canceled before the end of the trading day, reported in milliseconds and as the median across all stocks as well as members of each trader group; *Liquidity supply* is the fraction of the group's trading volume in all stocks where the member is on the passive side, counted in either SEK trading volume or number of trades. For each variable, the table also reports the p-value of the Wilcoxon rank sum z-test for equality between medians. As there are 23 (21) trading days in August 2011 (February 2012), and 30 stocks, the number of observations in each z-test is 690 (630).

Table 4: Market-making and opportunistic HFT activity across subsets of stocks

## Panel A: Market-making HFT activity

| Subsetting variable         | Tercile | Market-maker presence | Fraction of HFT trading volume | Fraction of HFT order submissions | Inventory | $q / t$ | Minimum latency (ms) | Limit order duration (ms) | Liquidity supply (volume) | Liquidity supply (trades) |
|-----------------------------|---------|-----------------------|--------------------------------|-----------------------------------|-----------|---------|----------------------|---------------------------|---------------------------|---------------------------|
| Market capitalization (SEK) | Higher  | 0.64                  | 0.63                           | 0.83                              | 0.08      | 9.25    | 0.08                 | 7568                      | 0.71                      | 0.69                      |
|                             | Lower   | 0.48                  | 0.69                           | 0.90                              | 0.15      | 12.69   | 0.09                 | 12461                     | 0.71                      | 0.71                      |
|                             | z-test  | 0.000                 | 0.000                          | 0.000                             | 0.000     | 0.000   | 0.000                | 0.000                     | 0.881                     | 0.294                     |
| Trading volume (SEK)        | Higher  | 0.65                  | 0.63                           | 0.83                              | 0.08      | 8.88    | 0.08                 | 7981                      | 0.71                      | 0.69                      |
|                             | Lower   | 0.52                  | 0.66                           | 0.87                              | 0.17      | 14.11   | 0.09                 | 13767                     | 0.69                      | 0.68                      |
|                             | z-test  | 0.000                 | 0.026                          | 0.000                             | 0.000     | 0.000   | 0.000                | 0.000                     | 0.318                     | 0.803                     |
| Realized volatility         | Higher  | 0.50                  | 0.63                           | 0.87                              | 0.13      | 11.72   | 0.09                 | 10324                     | 0.71                      | 0.72                      |
|                             | Lower   | 0.62                  | 0.62                           | 0.87                              | 0.13      | 10.13   | 0.09                 | 12310                     | 0.69                      | 0.69                      |
|                             | z-test  | 0.000                 | 0.765                          | 0.132                             | 0.283     | 0.004   | 0.369                | 0.020                     | 0.160                     | 0.139                     |
| Relative bid-ask spread     | Higher  | 0.65                  | 0.69                           | 0.87                              | 0.10      | 10.06   | 0.09                 | 9876                      | 0.69                      | 0.68                      |
|                             | Lower   | 0.54                  | 0.58                           | 0.85                              | 0.12      | 12.48   | 0.09                 | 9121                      | 0.72                      | 0.72                      |
|                             | z-test  | 0.000                 | 0.000                          | 0.018                             | 0.132     | 0.000   | 0.055                | 0.330                     | 0.020                     | 0.055                     |

Panel B: Opportunistic HFT activity

| Subsetting variable         | Tercile | Market-maker presence | Fraction of HFT trading volume | Fraction of HFT order submissions | Inventory | $q / t$ | Minimum latency (ms) | Limit order duration (ms) | Liquidity supply (volume) | Liquidity supply (trades) |
|-----------------------------|---------|-----------------------|--------------------------------|-----------------------------------|-----------|---------|----------------------|---------------------------|---------------------------|---------------------------|
| Market capitalization (SEK) | Higher  | 0.09                  | 0.37                           | 0.17                              | 0.51      | 4.29    | 0.27                 | 4517                      | 0.37                      | 0.36                      |
|                             | Lower   | 0.01                  | 0.31                           | 0.10                              | 0.48      | 4.18    | 0.84                 | 2246                      | 0.19                      | 0.19                      |
|                             | z-test  | 0.000                 | 0.000                          | 0.000                             | 0.202     | 0.108   | 0.000                | 0.003                     | 0.000                     | 0.000                     |
| Trading volume (SEK)        | Higher  | 0.09                  | 0.37                           | 0.17                              | 0.49      | 4.20    | 0.28                 | 4685                      | 0.36                      | 0.36                      |
|                             | Lower   | 0.02                  | 0.34                           | 0.13                              | 0.51      | 5.05    | 0.58                 | 2750                      | 0.22                      | 0.20                      |
|                             | z-test  | 0.000                 | 0.026                          | 0.000                             | 0.271     | 0.005   | 0.000                | 0.029                     | 0.000                     | 0.000                     |
| Realized volatility         | Higher  | 0.03                  | 0.37                           | 0.13                              | 0.46      | 3.71    | 0.46                 | 2376                      | 0.23                      | 0.22                      |
|                             | Lower   | 0.05                  | 0.38                           | 0.13                              | 0.54      | 3.66    | 0.67                 | 5719                      | 0.32                      | 0.29                      |
|                             | z-test  | 0.000                 | 0.765                          | 0.132                             | 0.000     | 0.690   | 0.439                | 0.000                     | 0.000                     | 0.000                     |
| Relative bid-ask spread     | Higher  | 0.05                  | 0.31                           | 0.13                              | 0.51      | 4.55    | 0.45                 | 3302                      | 0.38                      | 0.32                      |
|                             | Lower   | 0.06                  | 0.42                           | 0.15                              | 0.49      | 4.60    | 0.28                 | 2925                      | 0.30                      | 0.30                      |
|                             | z-test  | 0.000                 | 0.000                          | 0.018                             | 0.382     | 0.980   | 0.000                | 0.709                     | 0.001                     | 0.550                     |

Table 4 presents the medians of the trading and quoting activity for subsets of OMXS 30 index stocks during February 2012. Results for market-making HFTs are given in Panel A, and results for opportunistic HFTs are in Panel B. The subsetting variables are market capitalization (SEK), trading volume (SEK), realized volatility, and the relative bid-ask spread of the previous month (July 2011 and January 2012, respectively). Stocks are sorted each month and time series medians are reported for stocks with the 10 highest and 10 lowest values of each subsetting variable for the previous month. Each underlying statistic is at a daily frequency, where the daily observations are calculated as follows: *Market-maker presence* is the fraction of 10-second periods in the continuous trading session during which a member has quotes at either of the best bid and offer prices, averaged across the members in each group (market-making HFTs and opportunistic HFTs) and across all stocks; the *Fraction of HFT trading volume* is the sum of SEK volume traded, for all members in each group in all stocks, divided by the total HFT SEK trading volume in all stocks; the *Fraction of HFT order submissions* is the sum of order submissions posted for all stocks by the members in each group, divided by the total number of HFT order submissions in all stocks; *|Inventory|* is the absolute accumulated inventory divided by the trading volume (in number of shares), reported as the median across the members in each group and across all stocks; *q/t* is the aggregate number of add orders in all stocks divided by the aggregate number of executions in all stocks, for each group; *Minimum latency (Limit order duration)* is the minimum (median) lifetime of limit orders that are canceled before the end of the trading day, reported in milliseconds and as the median across all stocks as well as across the members of each trader group; *Liquidity supply* is the fraction of the group's trading volume in all stocks where the member is on the passive side, counted in either SEK trading volume or number of trades. For each variable, the table also reports the p-value of the Wilcoxon rank sum z-test for equality between medians. As there are 23 (21) trading days in August 2011 (February 2012), and 10 stocks in each tercile, the number of observations in each statistical test is 230 (210).

Table 5: Calculating HFTs' abnormal trading volume market share following tick size changes

|                        |                    | Event stocks |       |          | Benchmark stocks |       |          | Abnormal change |
|------------------------|--------------------|--------------|-------|----------|------------------|-------|----------|-----------------|
|                        |                    | Low          | High  | High/Low | Low              | High  | High/Low |                 |
| Before January 1, 2011 | All HFTs           | 0.246        | 0.132 | 0.567    | 0.188            | 0.177 | 0.987    | -0.435          |
| After January 1, 2011  | All HFTs           | 0.416        | 0.361 | 0.889    | 0.430            | 0.412 | 0.978    | -0.036          |
|                        | Opportunistic HFTs | 0.173        | 0.101 | 0.587    | 0.149            | 0.135 | 0.997    | -0.414          |
|                        | Market-making HFTs | 0.203        | 0.261 | 1.169    | 0.308            | 0.281 | 0.965    | 0.161           |

Table 5 shows statistics used for calculating abnormal effects of tick size changes on HFT SEK trading volumes. Trading volume market share for trader group  $j = [All\ HFTs, Market-making\ HFTs, Opportunistic\ HFTs]$  is defined as  $\theta_{j,t}^{vol} = tvol_{j,t}/tvol_t$ , where  $tvol_t$  is the sum of HFT and Non-HFT trading volumes on day  $t$ , and  $tvol_{j,t}$  is the trader group  $j$  trading volume on day  $t$ . For each tick size change, trading volume market shares are calculated for each group on the closest trading day where the lower tick size holds throughout the day (Low), and the closest trading day where the high tick size holds throughout the day (High). The ratio of the High and Low statistics (High/Low) describes the tick size event effect. The median of Low, High, and High/Low are reported for both the stocks experiencing the tick size change (Event stocks), and stocks in the OMXS30 that do not have a concurrent tick size change (Benchmark stocks). The abnormal market share change, for which the median of each trader group is reported, is defined as the difference in High/Low ratios of the Event stocks and the Benchmark stocks. The analysis uses 89 events during the sample period, whereof 42 observations are before January 1, 2011, and 47 after January 1, 2011.

Table 6: Abnormal HFT market share changes associated with tick size changes

|                        |                    | Trading volume     |                    |                    | Number of executions |                    |                    | Number of limit orders |                    |
|------------------------|--------------------|--------------------|--------------------|--------------------|----------------------|--------------------|--------------------|------------------------|--------------------|
|                        |                    | Total              | Active             | Passive            | Total                | Active             | Passive            | Submissions            | Cancellations      |
| Before January 1, 2011 | All HFTs           | -0.435<br>(0.0000) | -0.475<br>(0.0000) | -0.164<br>(0.5822) | -0.443<br>(0.0000)   | -0.499<br>(0.0000) | -0.149<br>(0.5652) | -0.290<br>(0.0001)     | -0.248<br>(0.0003) |
| After January 1, 2011  | All HFTs           | -0.036<br>(0.3436) | -0.189<br>(0.0002) | 0.067<br>(0.0348)  | 0.033<br>(0.9367)    | -0.191<br>(0.0103) | 0.061<br>(0.0339)  | -0.023<br>(0.1161)     | -0.019<br>(0.2876) |
|                        | Opportunistic HFTs | -0.414<br>(0.0000) | -0.498<br>(0.0000) | -0.167<br>(0.2297) | -0.361<br>(0.0000)   | -0.473<br>(0.0000) | -0.047<br>(0.3656) | 0.008<br>(0.2176)      | 0.138<br>(0.0208)  |
|                        | Market-making HFTs | 0.161<br>(0.0008)  | 0.237<br>(0.0029)  | 0.129<br>(0.0046)  | 0.165<br>(0.0013)    | 0.173<br>(0.0214)  | 0.094<br>(0.0196)  | -0.046<br>(0.2339)     | -0.055<br>(0.1912) |

Table 6 contains medians for the abnormal effects of tick size changes on HFT activities: trading volume (Volume), executed transactions (Executions), limit order submissions (Submissions), and limit order cancellations (Cancellations) across events. Each abnormal effect is defined as HFT activity relative to the HFTs and non-HFTs activity combined, in a stock that experiences a change in tick size (Event stocks), in the High-Tick regime relative to the Low-Tick regime, relative the average corresponding fraction of activity in non-event stocks (Benchmark stocks). For example, abnormal trading volume market share for trader group  $j$  is:

$$\Delta \tilde{\theta}_j^{tvol} = [\tilde{\theta}_j^{tvol}]_{Event} - [\tilde{\theta}_j^{tvol}]_{Benchmark}$$

where  $\tilde{\theta}_j^{tvol} = \theta_{j,High}^{tvol} / \theta_{j,Low}^{tvol}$  is the market share change from the Low-Tick to the High-Tick regime,  $\theta_{j,t}^{tvol} = \frac{tvol_{j,t}}{tvol_t}$  is the market share of trader  $j = [All\ HFTs, Market-making\ HFTs, Opportunistic\ HFTs]$ , where  $tvol_t$  is the sum of HFT and Non-HFT trading volumes on day  $t$ , and  $tvol_{j,t}$  is the trader group  $j$  trading volume on day  $t$ . The table reports the median abnormal market share across events. Each median effect is tested using a Wilcoxon signed rank test, from which the  $p$ -value are displayed in parentheses. Active (passive) trading volume and number of executed transactions refers to the case when HFTs are initiating a market order or marketable limit order (are experiencing an execution of their standing limit order). The analysis uses 89 events during the sample period, whereof 42 observations are before January 1, 2011, and 47 after January 1, 2011.

Table 7: Intraday realized stock return volatility following tick size changes

| Volatility measure                                                     | 1 min              | 5 min              | 10 min             | 15 min             |
|------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|
| Panel A: Benchmarking to all non-event stocks                          |                    |                    |                    |                    |
| Before January 1, 2011                                                 | 0.116<br>(0.0016)  | 0.148<br>(0.0025)  | 0.132<br>(0.0027)  | 0.183<br>(0.0045)  |
| After January 1, 2011                                                  | 0.011<br>(0.3474)  | -0.055<br>(0.9304) | -0.051<br>(0.8016) | 0.009<br>(0.6860)  |
| After - Before                                                         | -0.104<br>(0.0542) | -0.203<br>(0.0152) | -0.182<br>(0.0100) | -0.174<br>(0.0773) |
| Panel B: Benchmarking to the non-event stock with similar event return |                    |                    |                    |                    |
| Before January 1, 2011                                                 | 0.203<br>(0.0003)  | 0.333<br>(0.0013)  | 0.145<br>(0.0133)  | 0.238<br>(0.0454)  |
| After January 1, 2011                                                  | -0.025<br>(0.6542) | 0.013<br>(0.7764)  | -0.027<br>(0.9477) | 0.070<br>(0.4982)  |
| After - Before                                                         | -0.227<br>(0.0069) | -0.319<br>(0.0214) | -0.172<br>(0.0585) | -0.168<br>(0.3430) |

Table 7 contains the medians for the abnormal effects of tick size changes on intraday realized stock return volatility, using midpoint quote changes in 1, 5, 10, and 15-minute intervals. Each abnormal volatility effect is defined as the volatility of a stock that experiences a change in tick size (Event stocks), in the High-Tick regime relative to the Low-Tick regime, relative the average corresponding volatility of non-event stocks (Benchmark stocks) according to:

$$\Delta\tilde{\sigma}(k) = \left[ \frac{\sigma(k)_{High}}{\sigma(k)_{Low}} \right]_{Event} - \left[ \frac{\sigma(k)_{High}}{\sigma(k)_{Low}} \right]_{Benchmark}$$

where  $\sigma(k)_{High}$  ( $\sigma(k)_{Low}$ ) is the volatility in the high (low) tick size regime, and  $k = 1, 5, 10$ , and 15 minutes. In Panel A, all OMXS30 that do not experience a tick size change at the time of the event are used as benchmark stocks. In Panel B, the stock in the OMXS30 that has an event return that matches the event return of the event stock most closely is used as Benchmark. Each panel reports median abnormal volatility across events. Each median effect is tested using a  $p$ -value from a Wilcoxon signed rank test, displayed in parentheses. In the sample period before January 1, 2011, only opportunistic HFTs are active. After January 1, 2011, both opportunistic HFTs and market making HFTs are active. After - Before denotes the difference between the median abnormal effects in these two periods. The analysis uses 89 events during the sample period, whereof 42 observations are before January 1, 2011, and 47 after January 1, 2011.